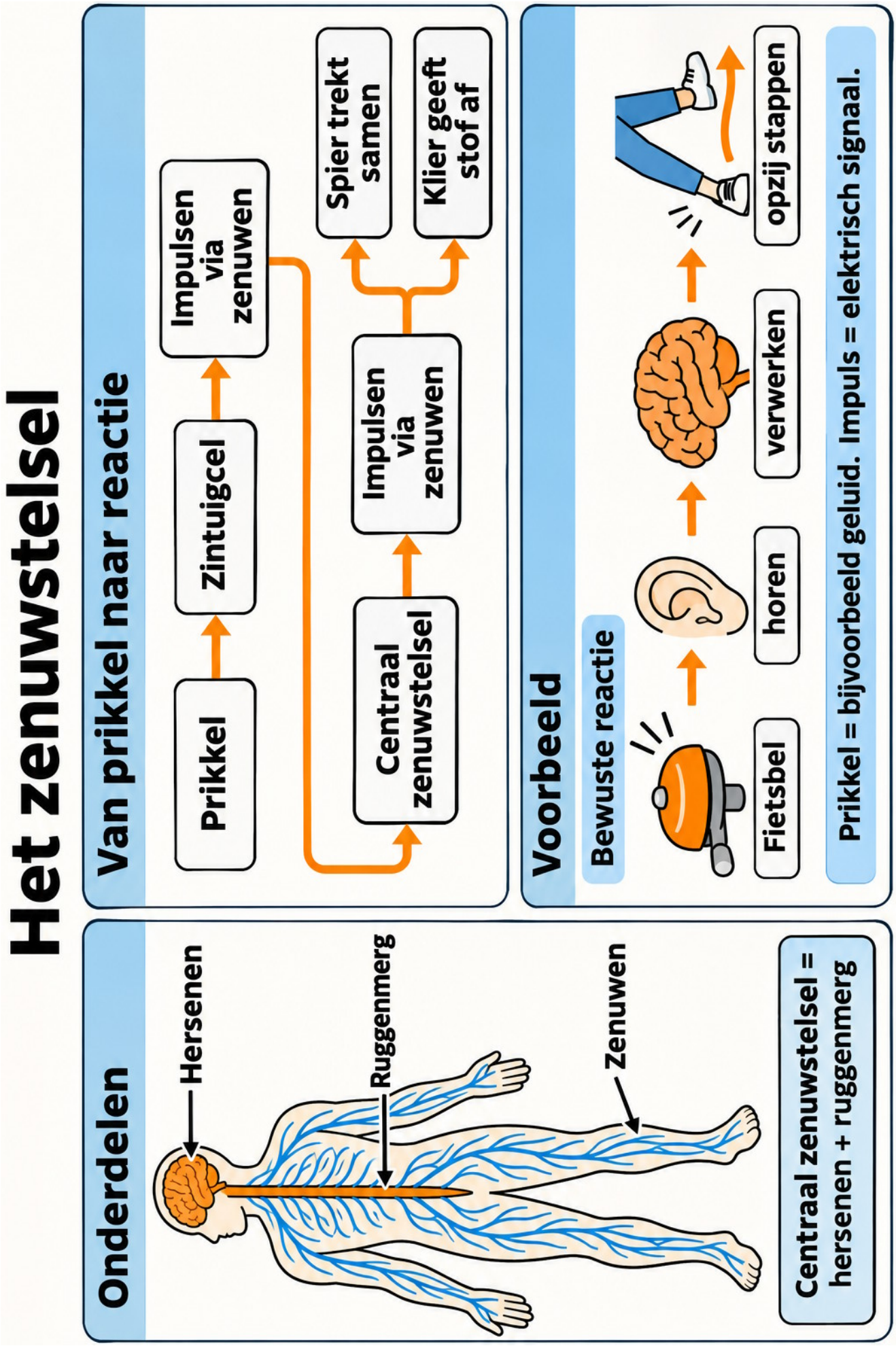
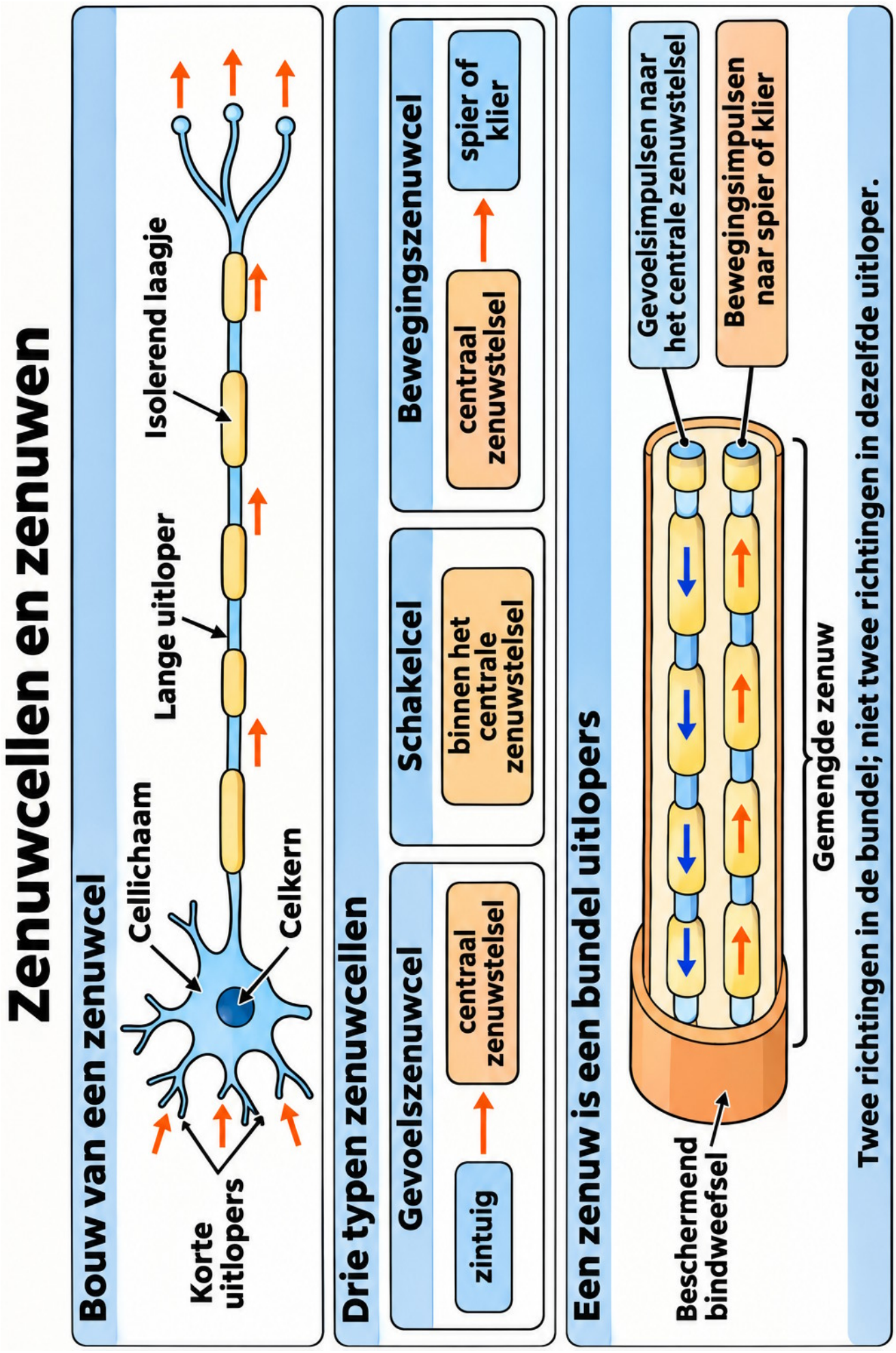
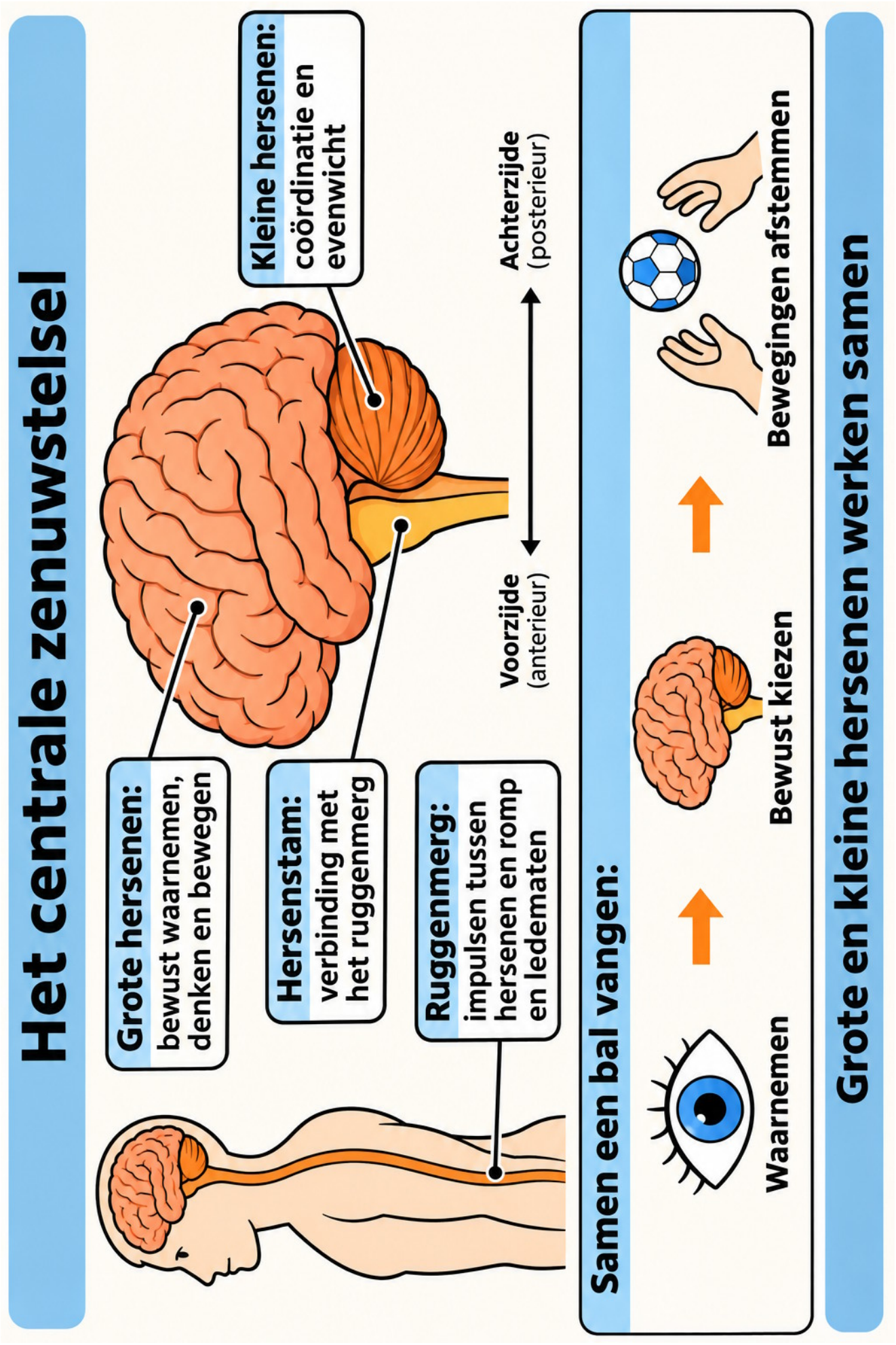
Basisstof 1 - Het zenuwstelsel



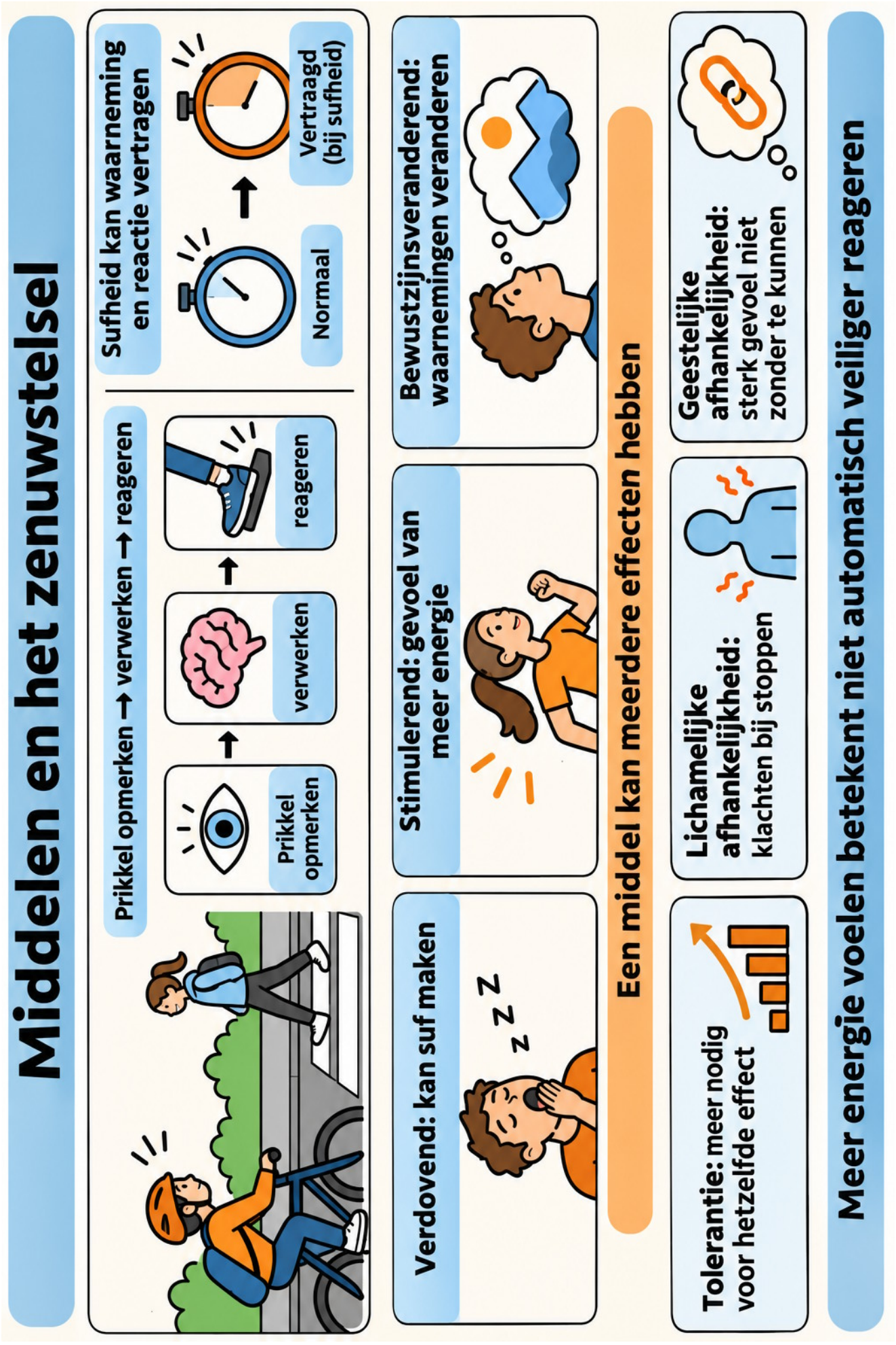
Basisstof 2 - Zenuwcellen en zenuwen



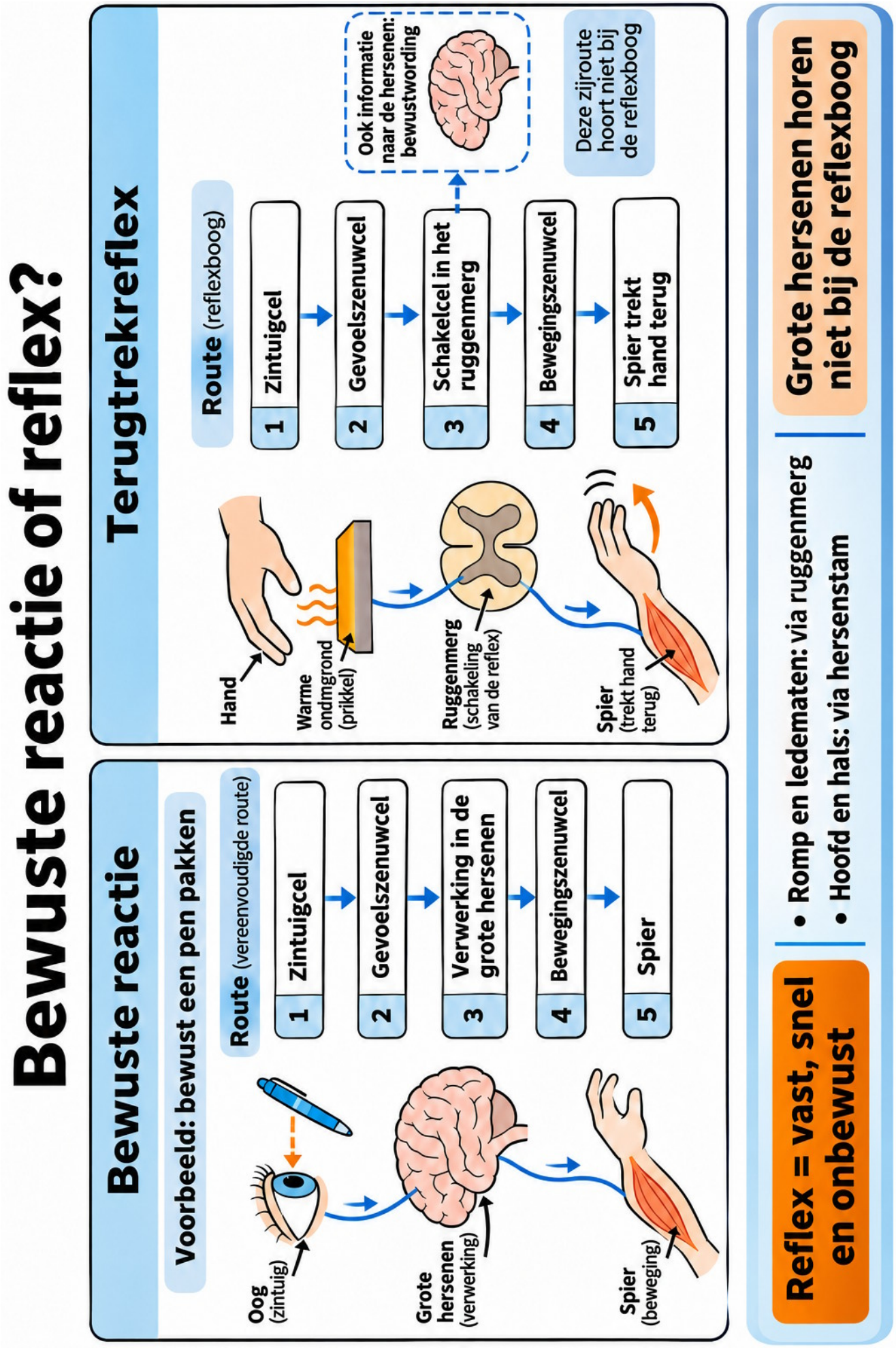
Basisstof 3 - Het centrale zenuwstelsel



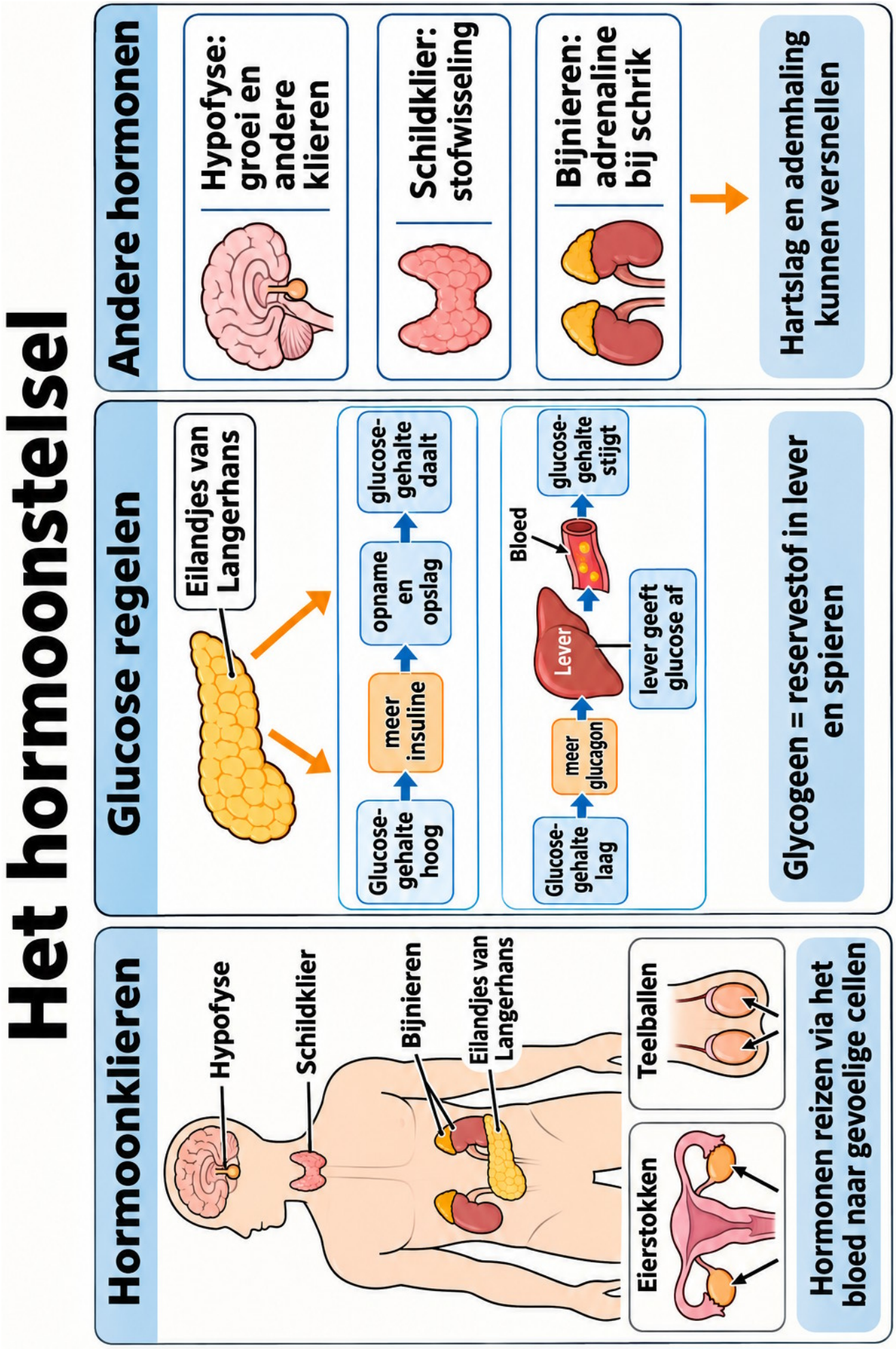
Basisstof 4 - Beïnvloeding van het zenuwstelsel



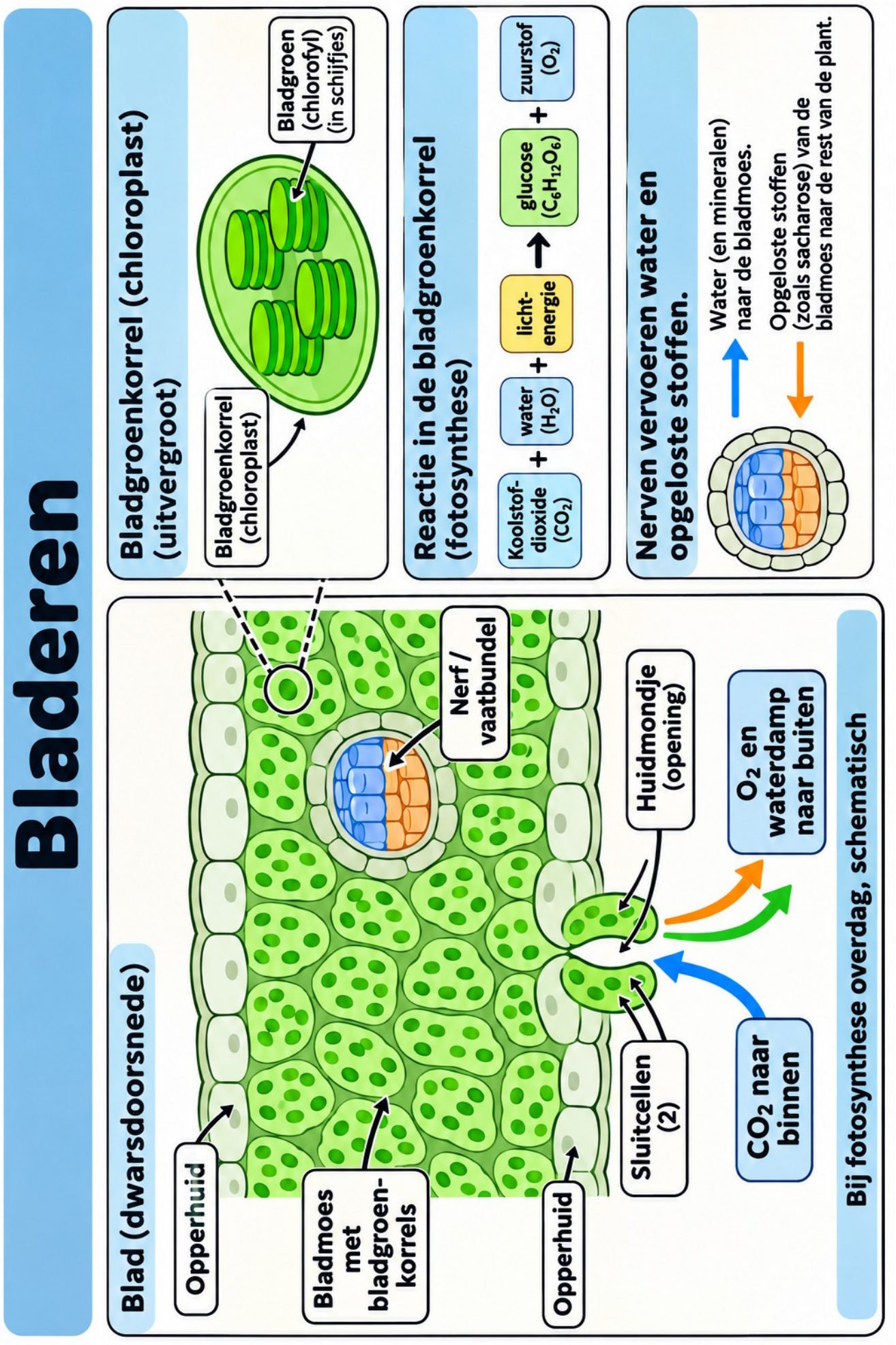
Basisstof 5 - Bewuste reacties en reflexen



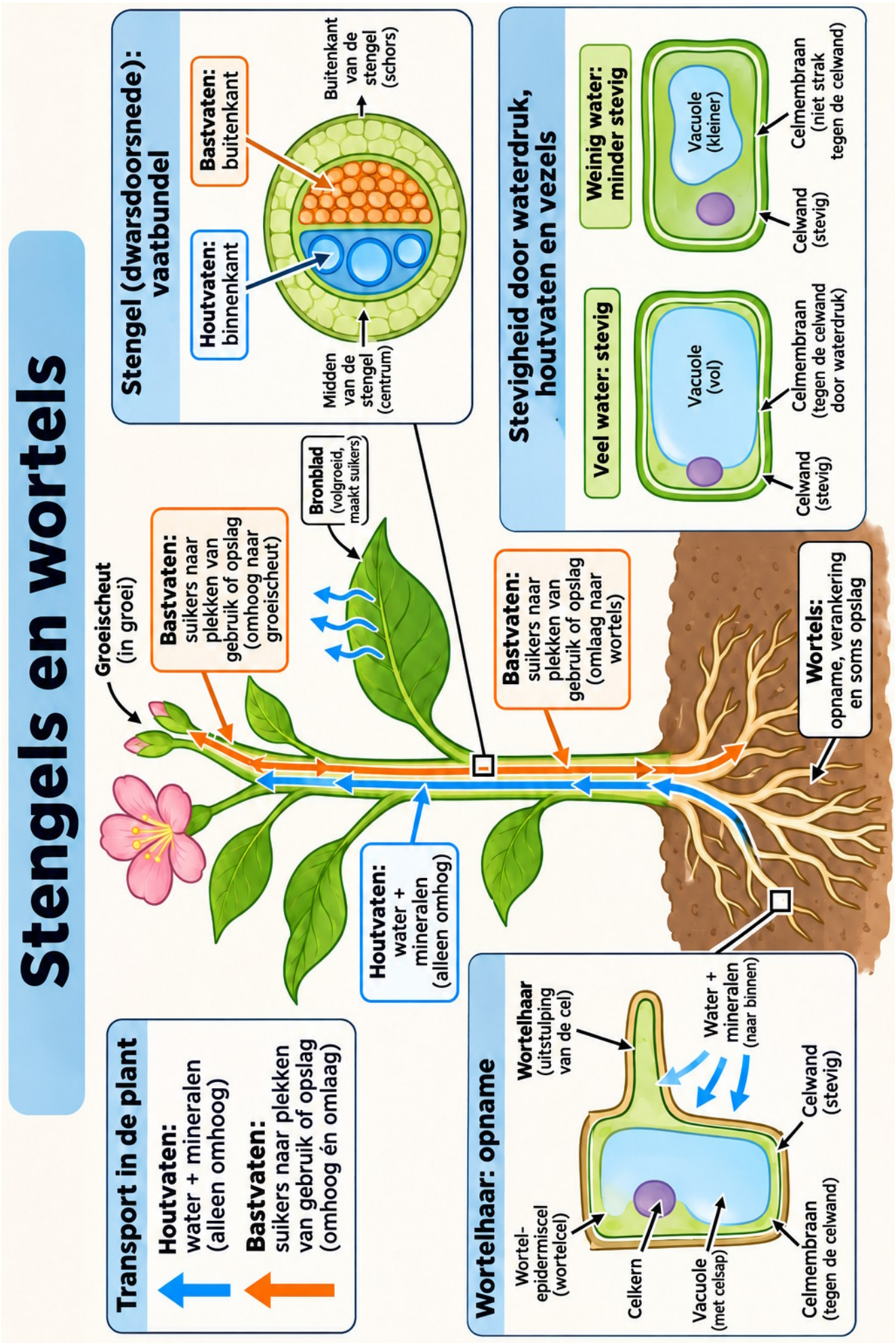
Basisstof 6 - Het hormoonstelsel



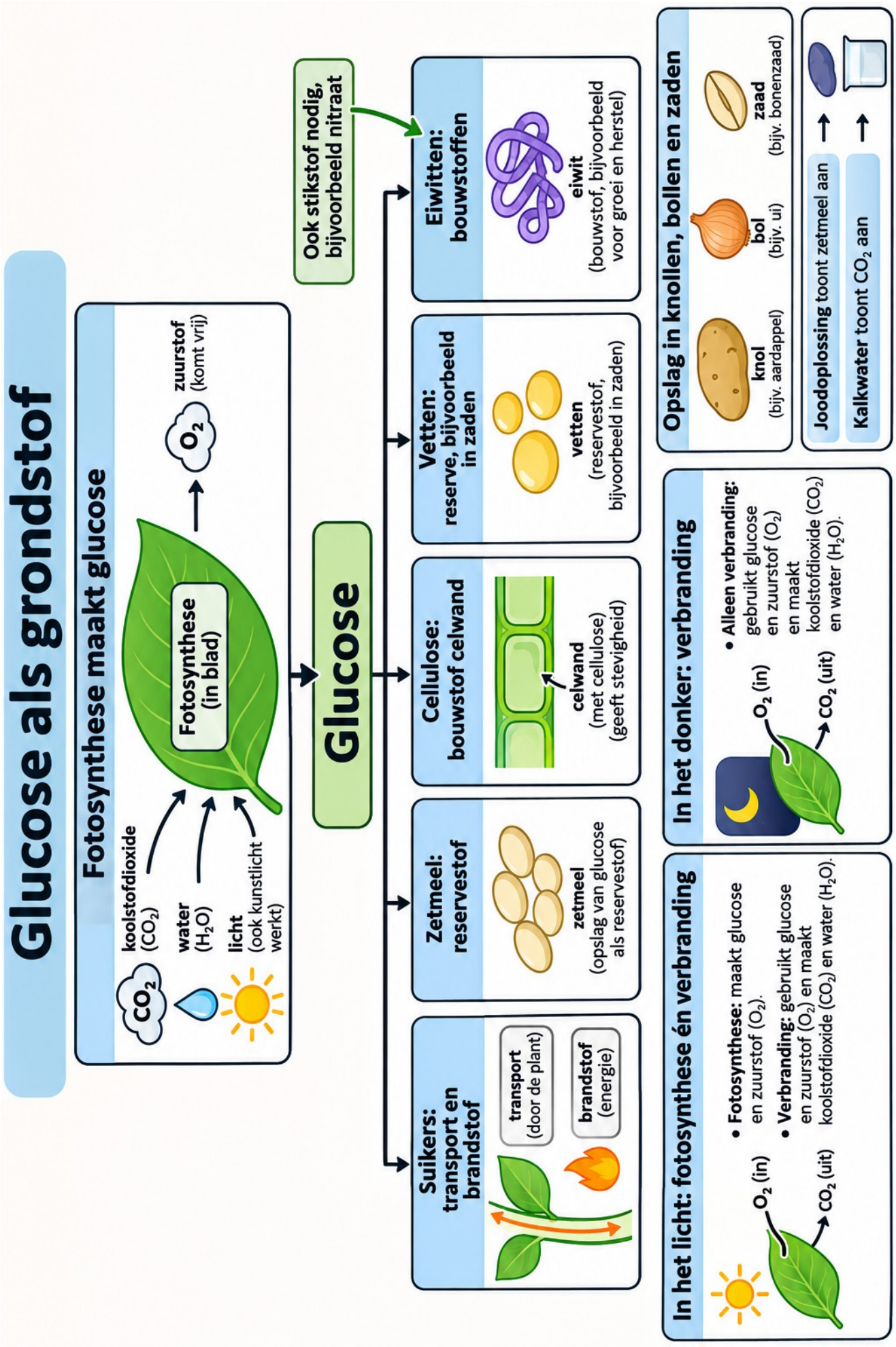
Basisstof 1, Bladeren



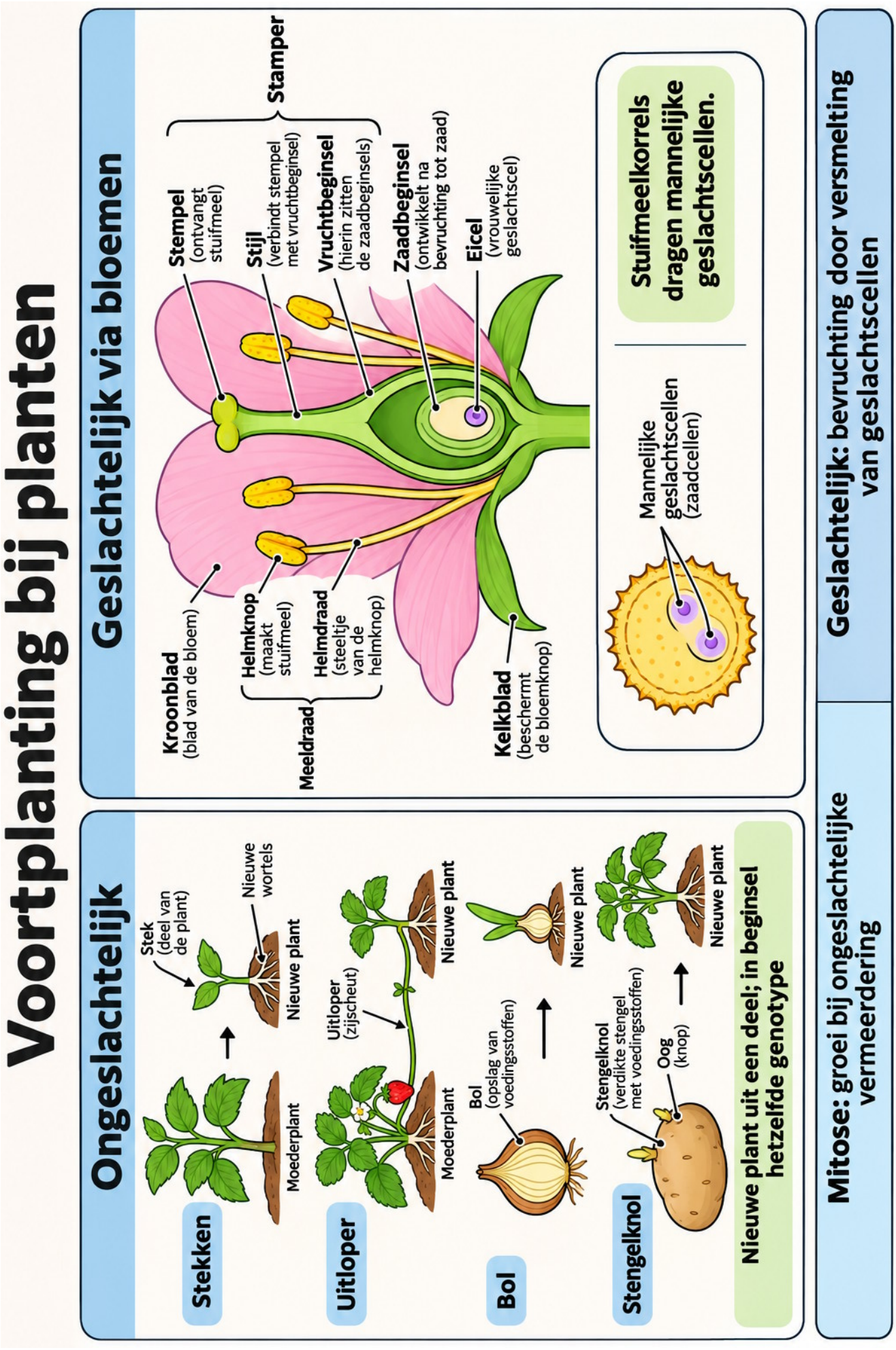
Basisstof 2, Stengels en wortels



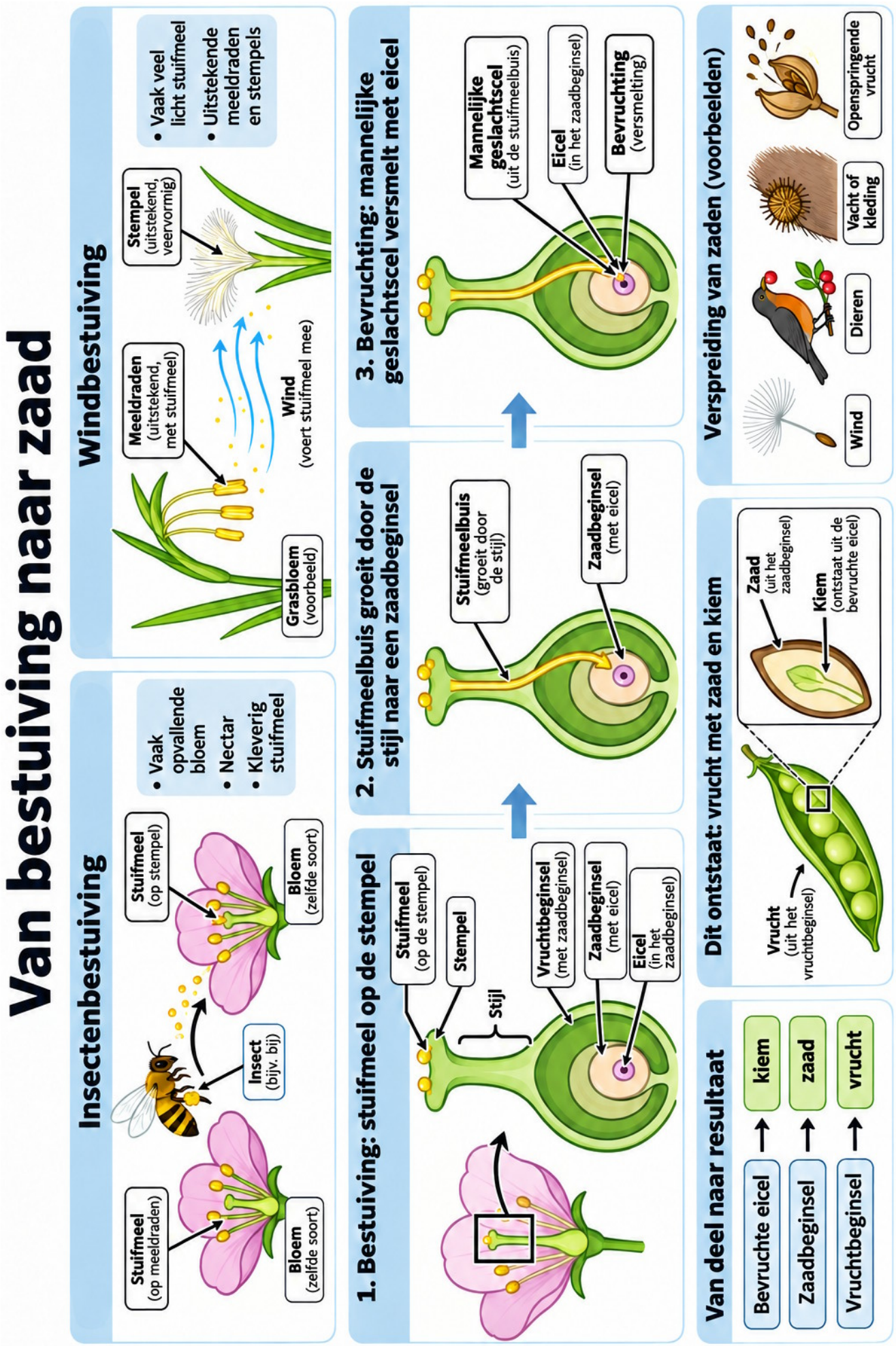
Basisstof 3, Glucose als grondstof



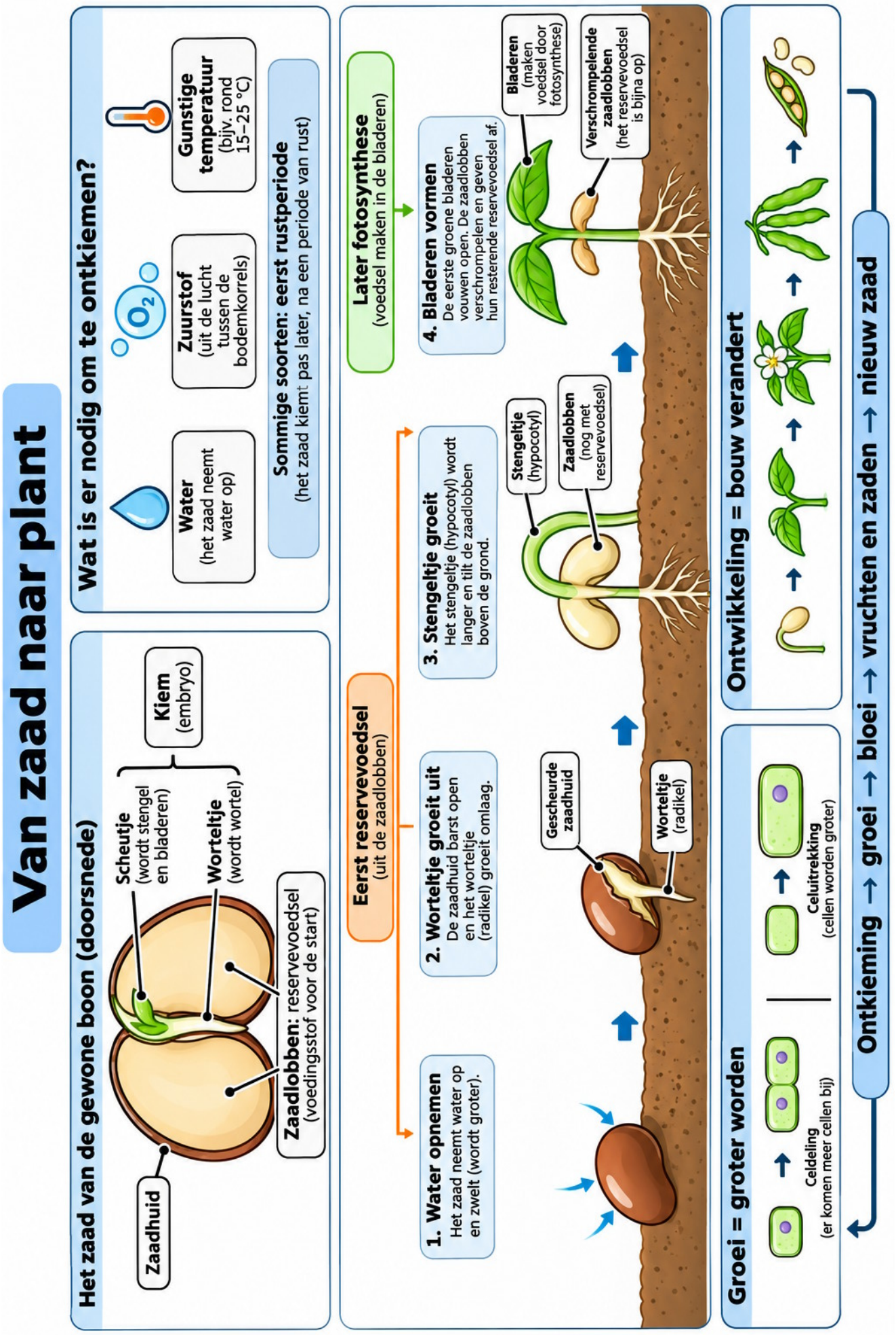
Basisstof 4, Voortplanting



Basisstof 5, Bestuiving en bevruchting

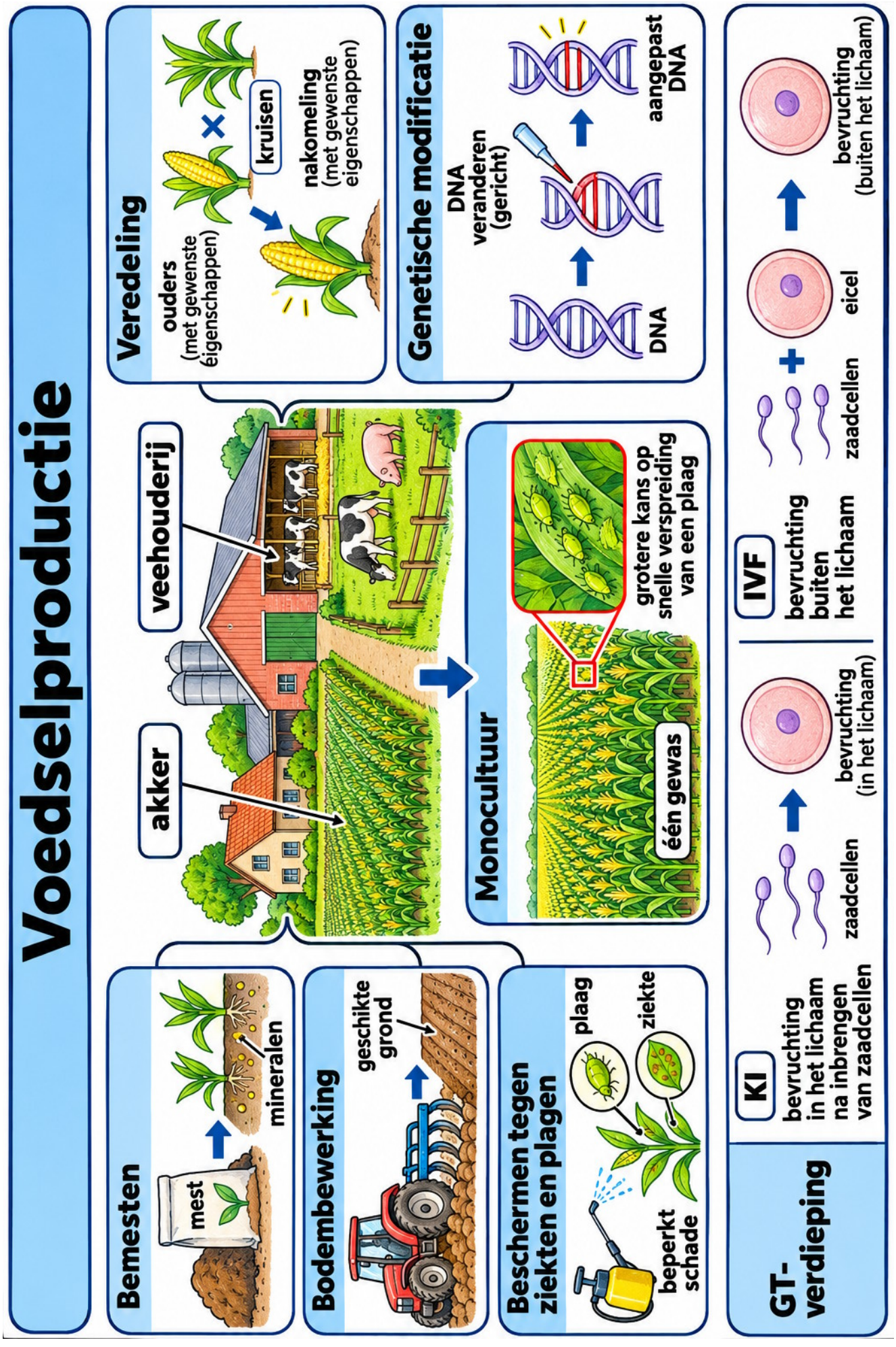


Basisstof 6, Ontkieming, groei en ontwikkeling

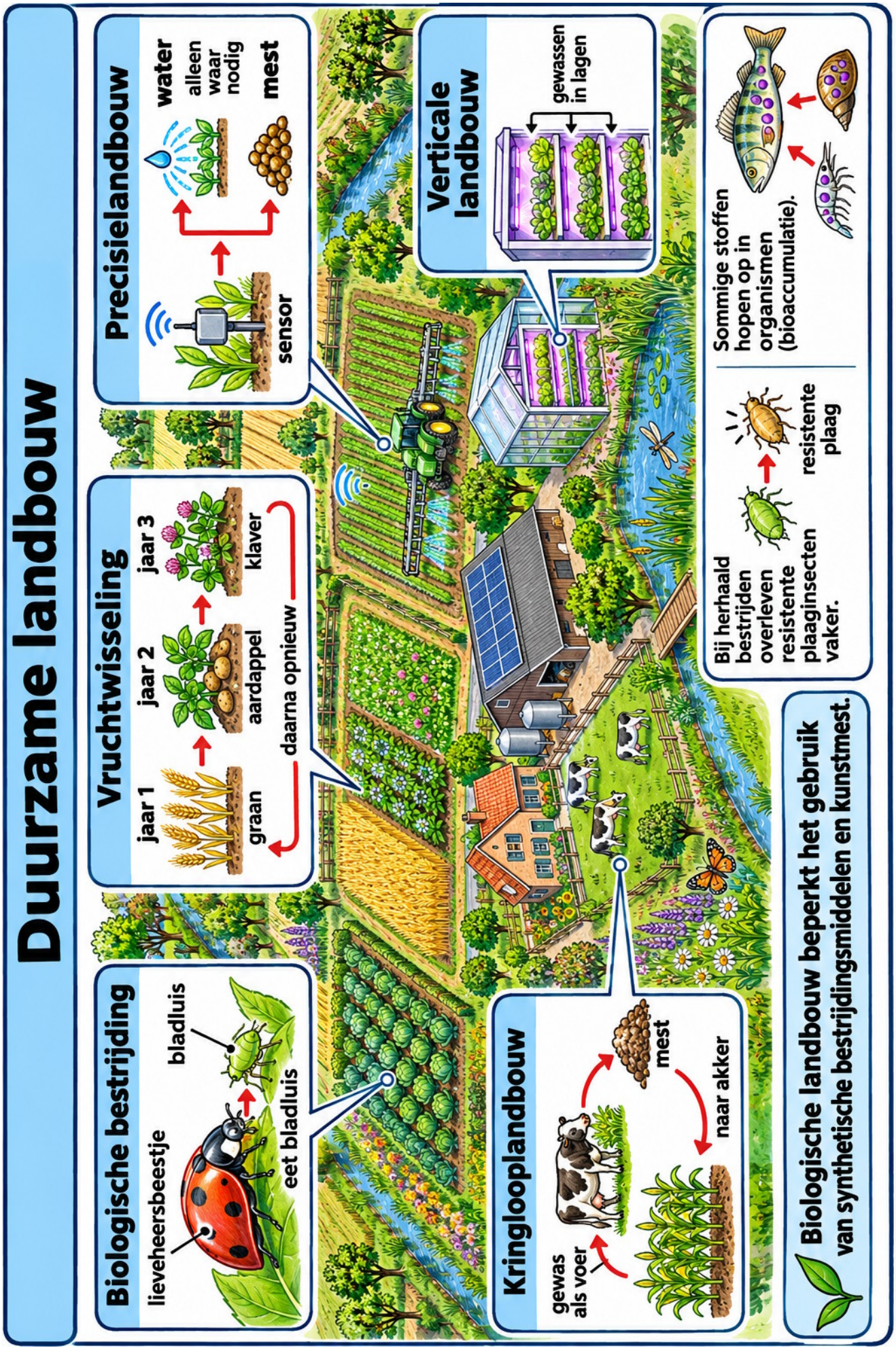


Basisstof 1, De mens en het milieu

Basisstof 2, Voedselproductie

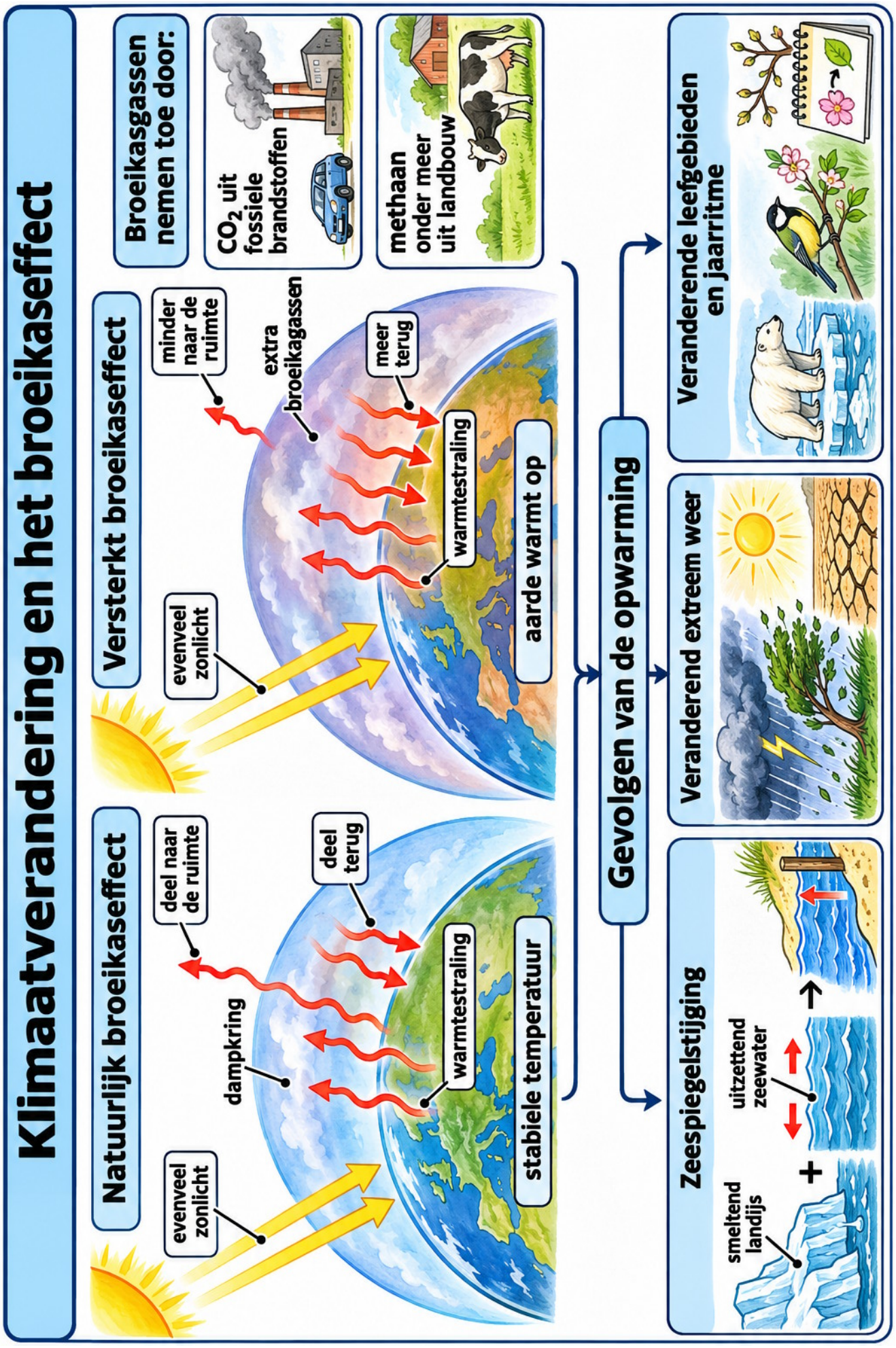


Basisstof 3, Duurzame landbouw

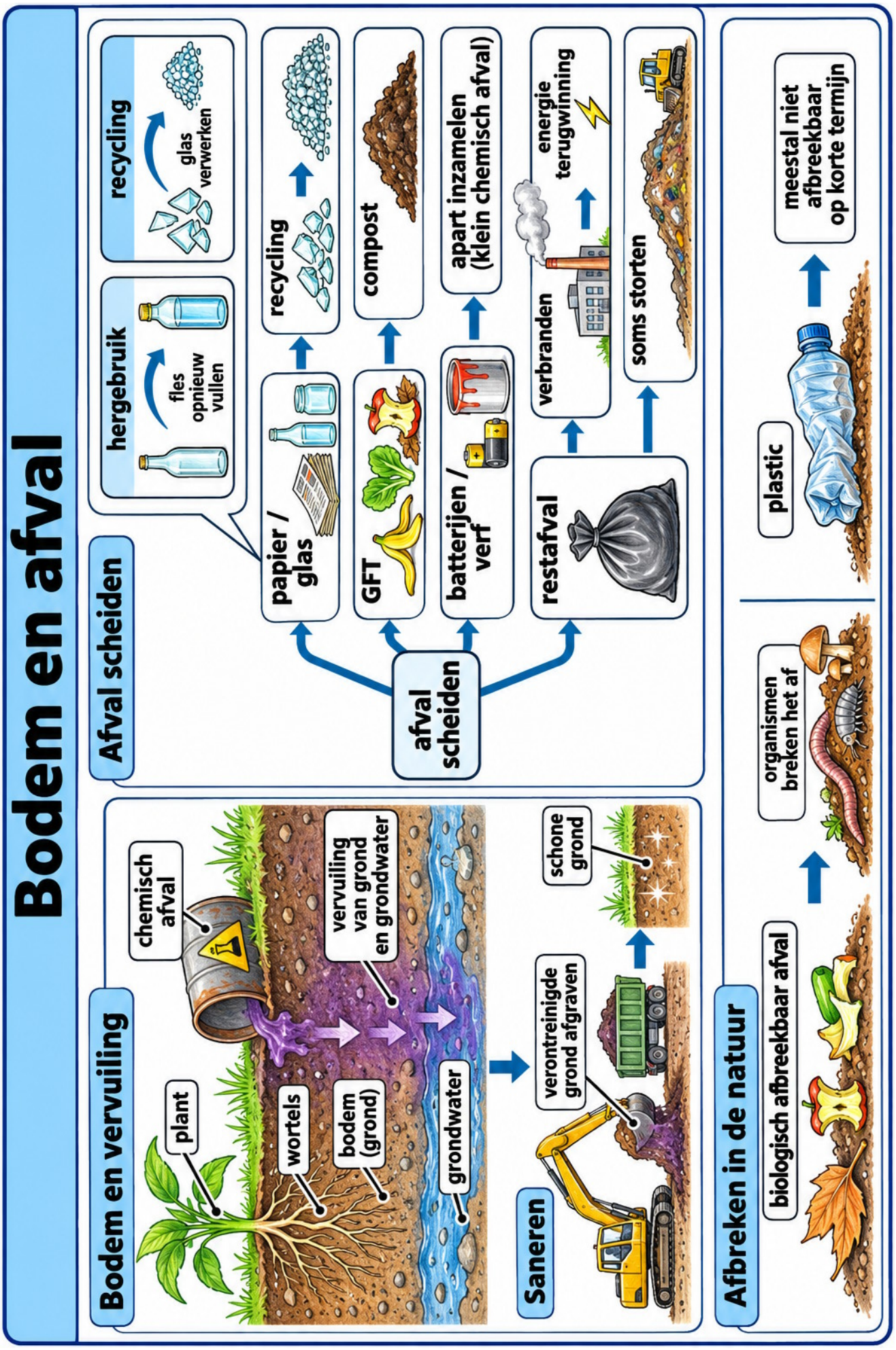


Basisstof 4, Energie

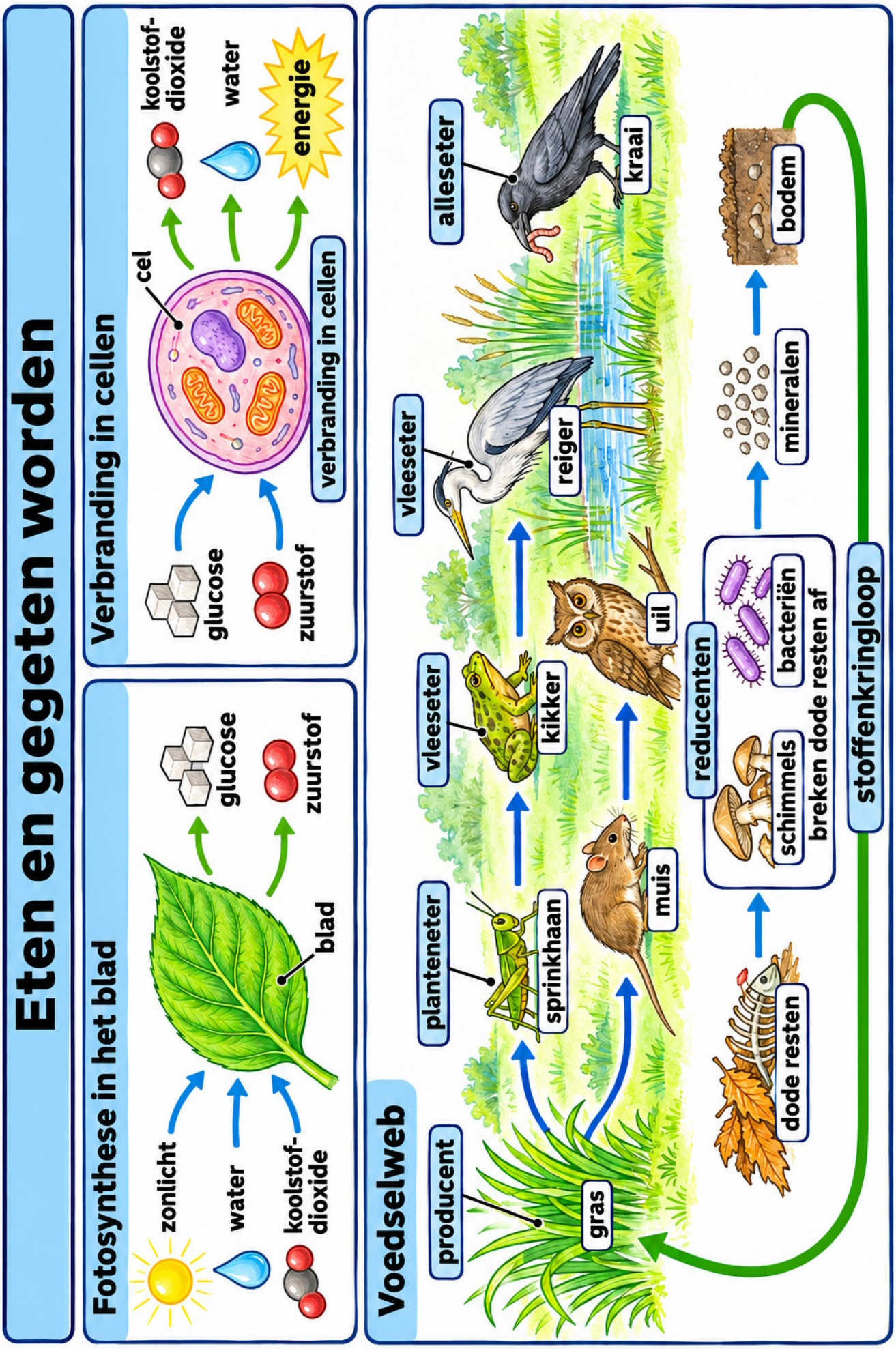
Basisstof 5, Klimaatverandering



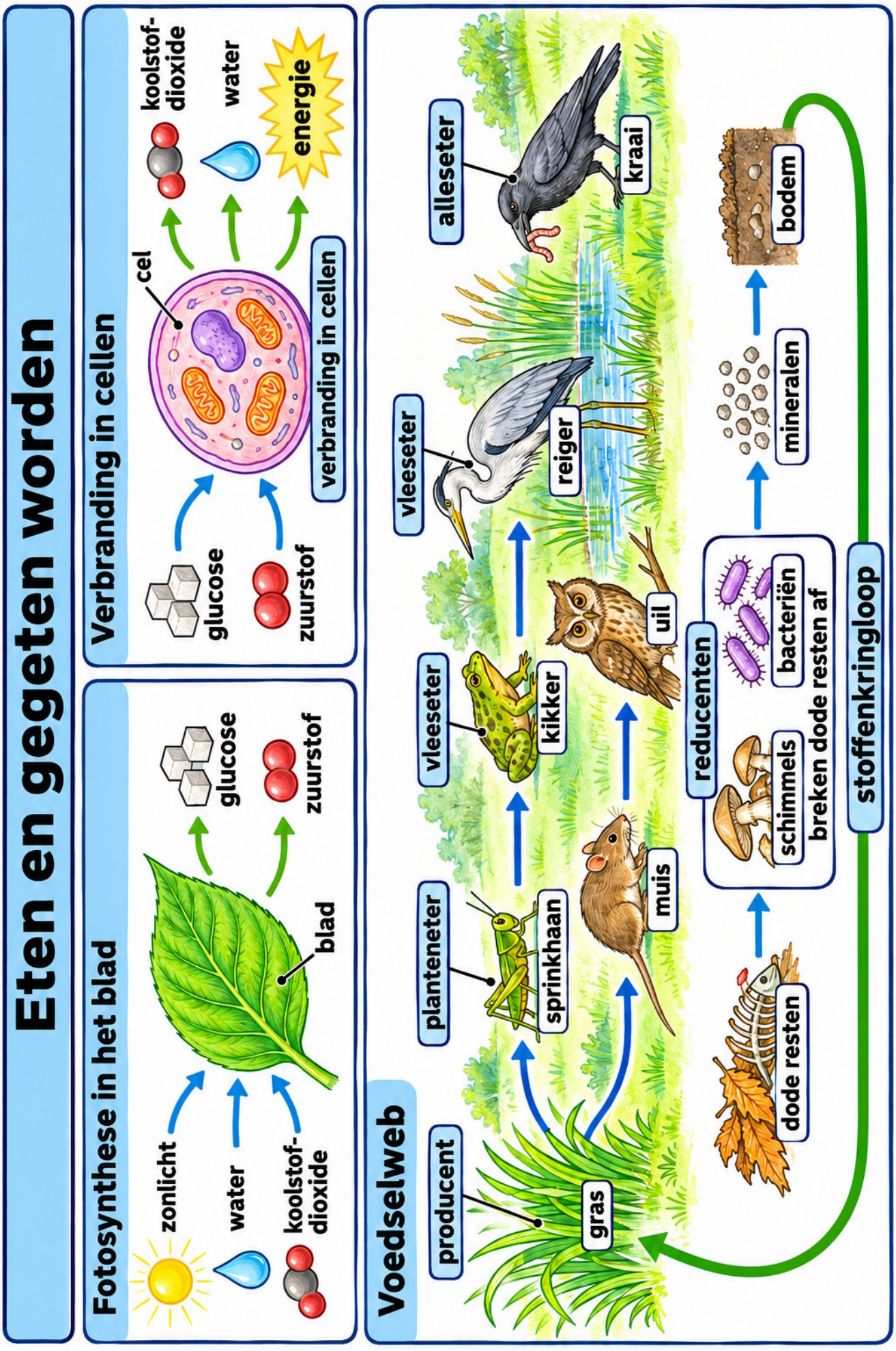
Basisstof 6, Water, bodem en afval



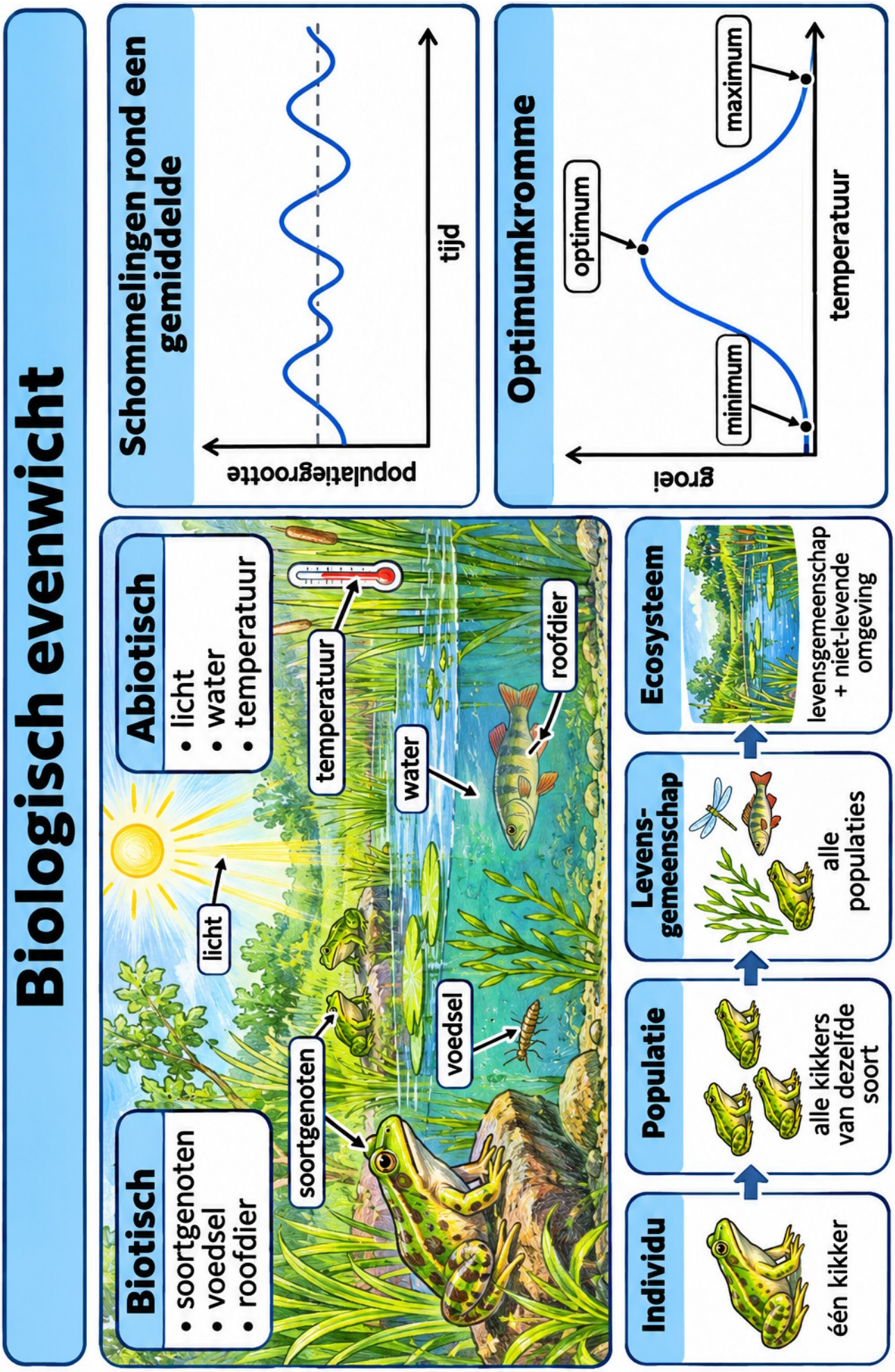
Basisstof 1, Fotosynthese en verbranding



Basisstof 2, Eten en gegeten worden

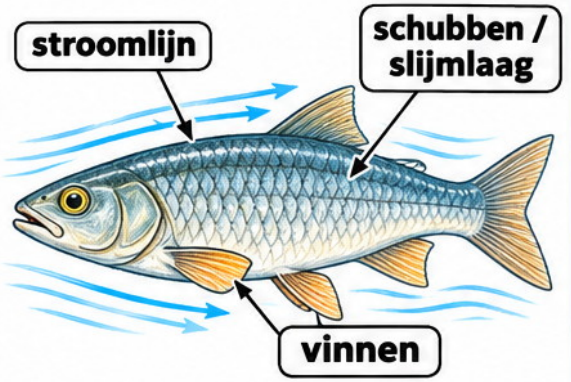


Basisstof 3, Organismen en hun leefomgeving

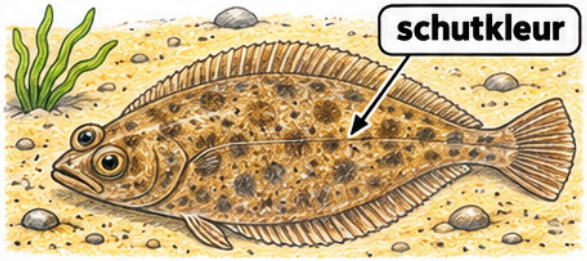


Aanpassingen bij dieren

Water



stroomlijn	minder weerstand in het water
vinnen	zwemmen, sturen en in evenwicht
schubben / slijmlaag	beschermen, minder wrijving



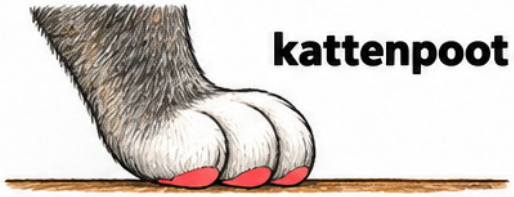
schutkleur	dezelfde kleur en tekening als de omgeving, zo valt hij niet op

Land

berenpoot

hele voetzool	veel contact met de grond, stabiel en sterk

kattenpoot



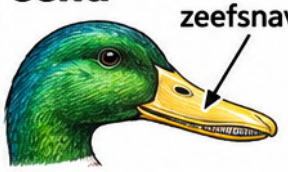
tenen	loopt op de tenen, soepel, stil en snel

paardenhoef

top van teen	loopt op de top van de teen, snel en efficiënt

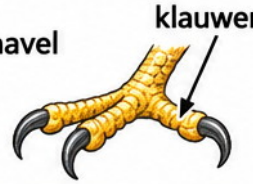
Vogels

eend



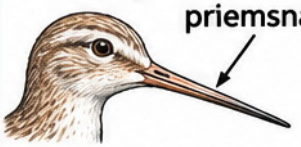
functie: voedsel uit het water zeven; goed zwemmen

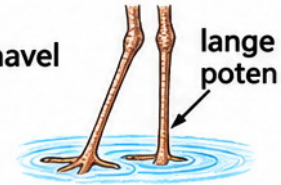
roofvogel



functie: prooi scheuren en vastgrijpen

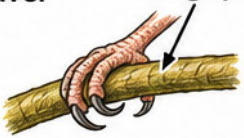
steltloper





functie: voedsel uit modder en ondiep water halen

vink

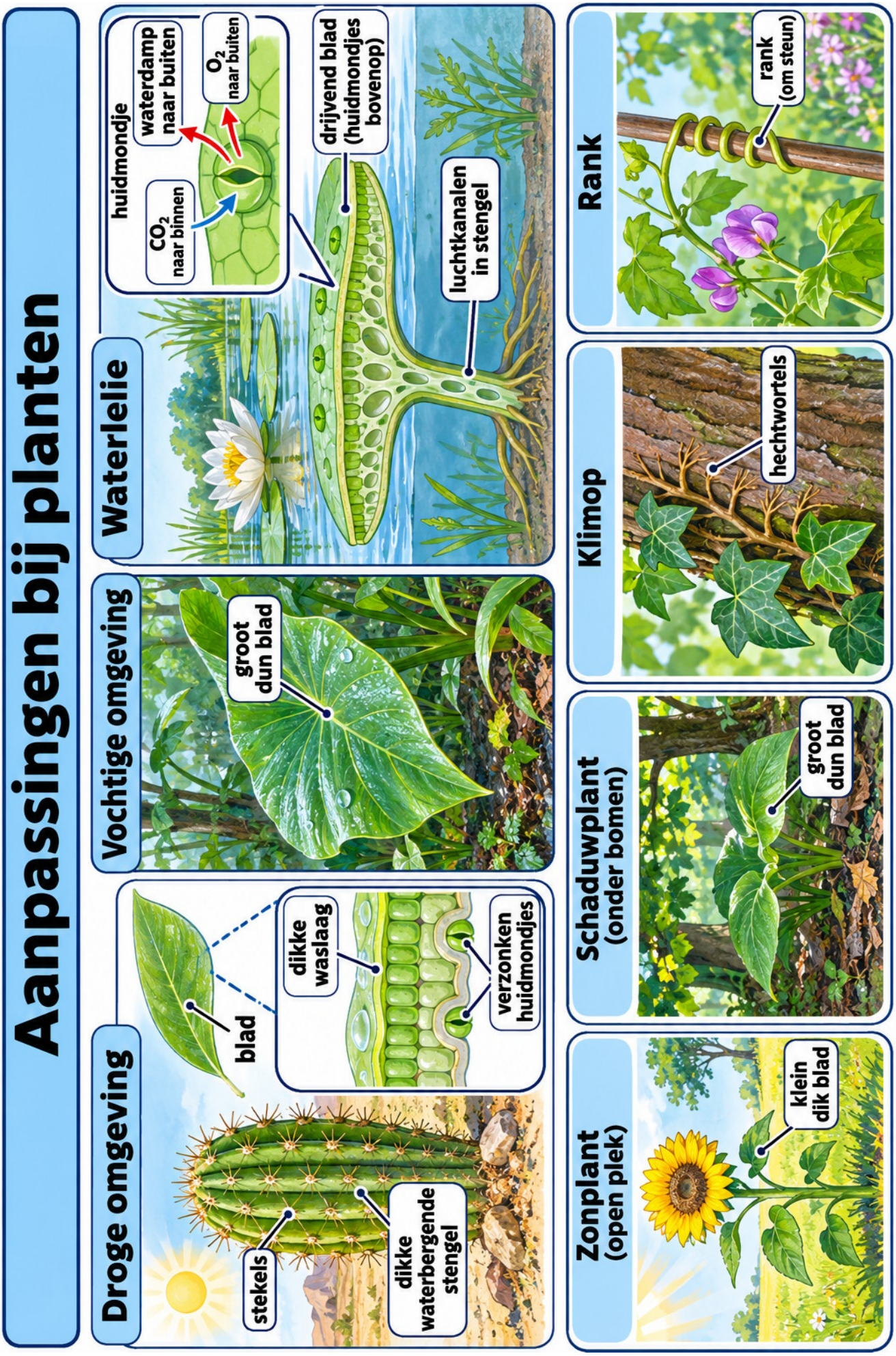


functie: zaden en pitten kraken; goed op takken zitten

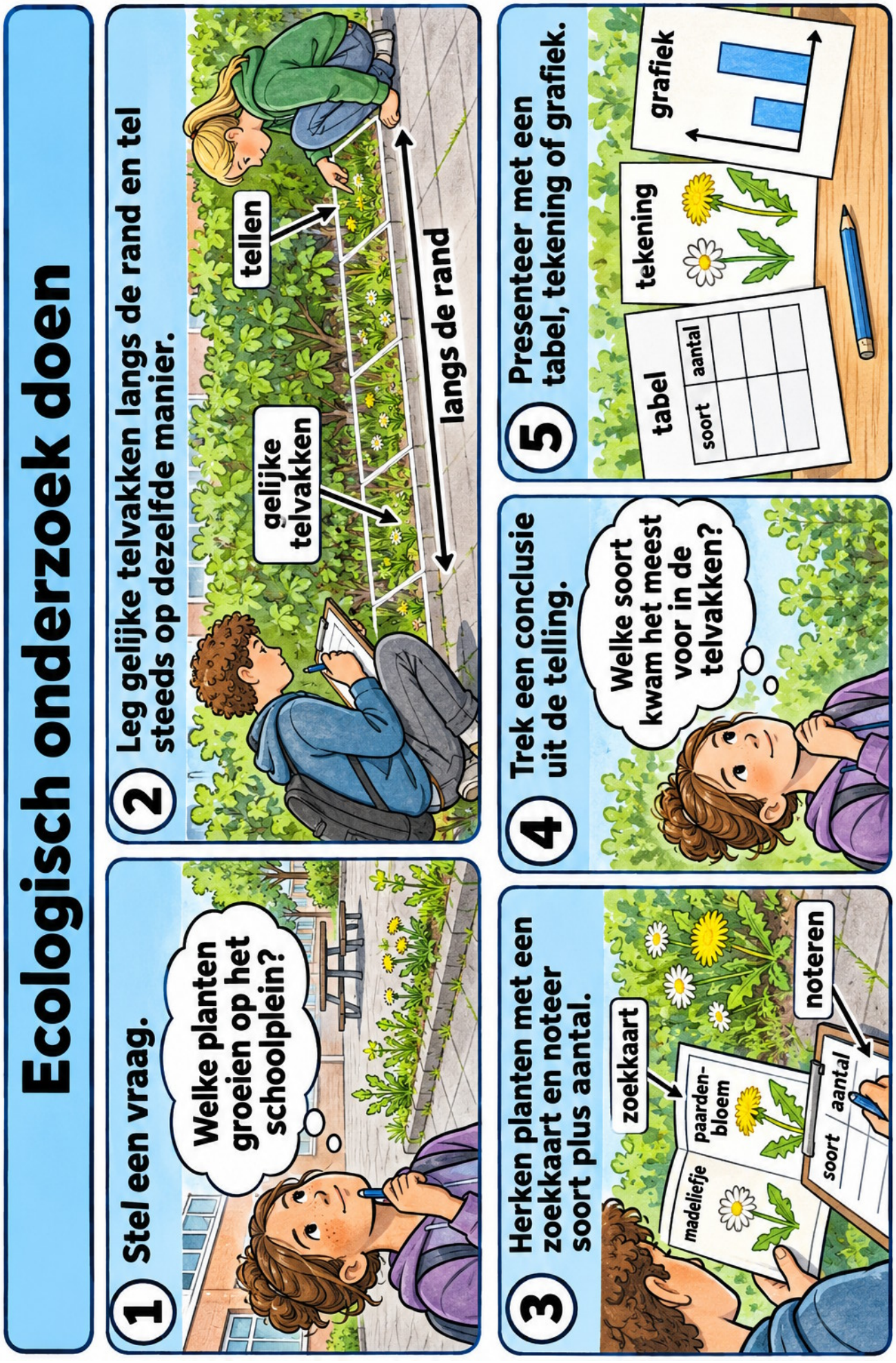
mees

functie: kleine diertjes (van bladeren) oppikken

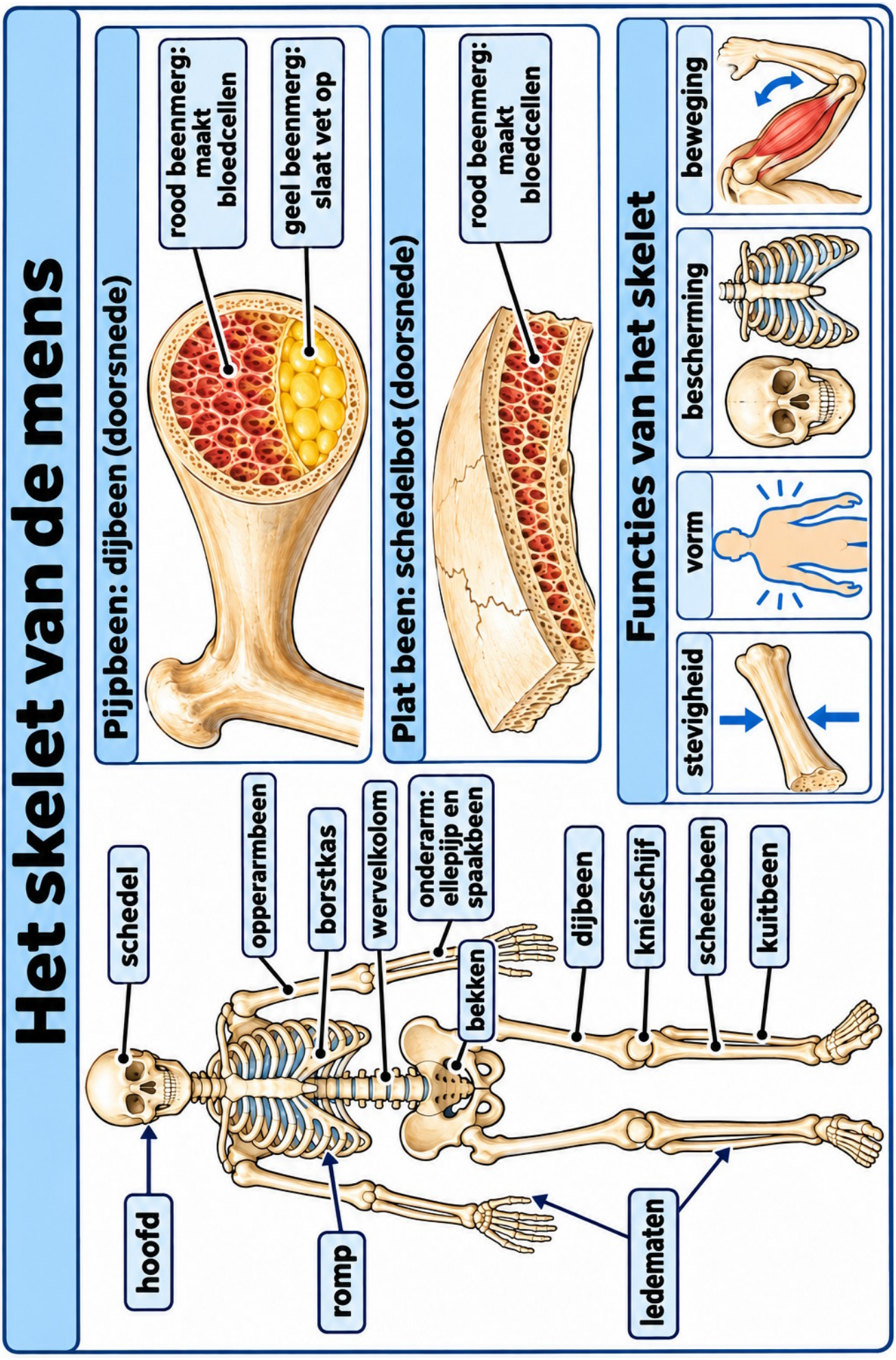
Basisstof 5, Aanpassingen bij planten



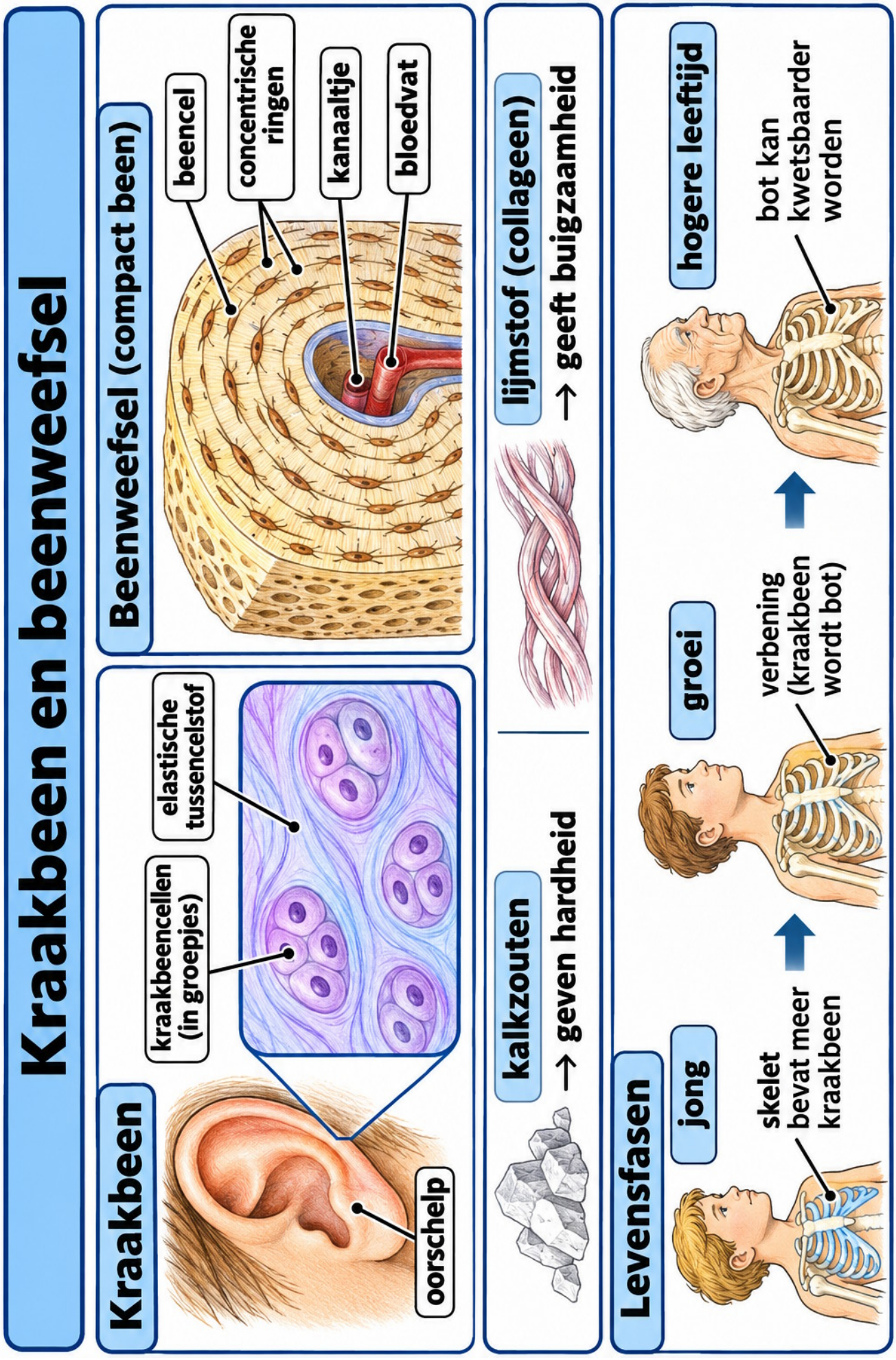
Basisstof 6, Ecologisch onderzoek doen



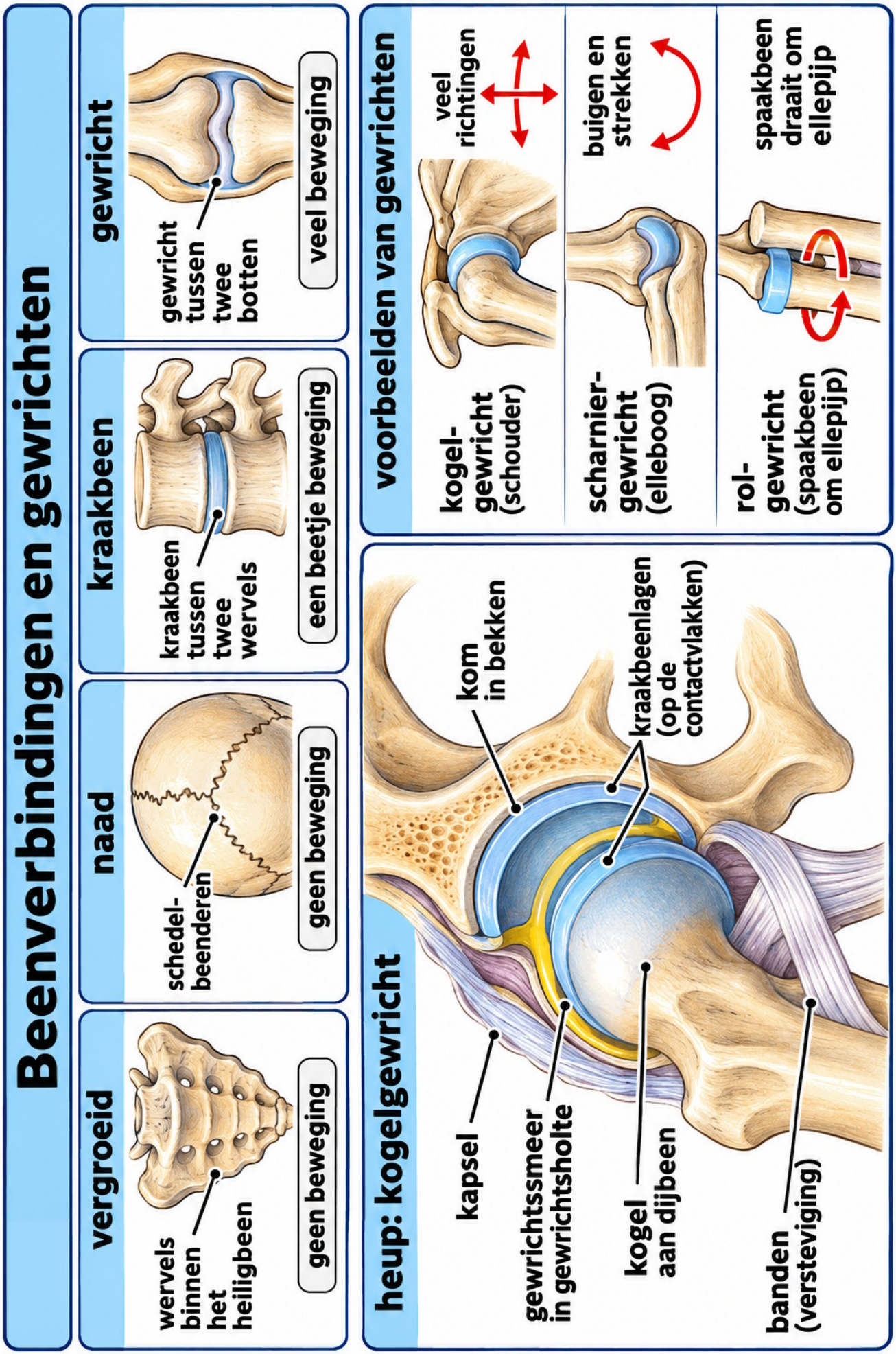
Basisstof 1, Het skelet van de mens



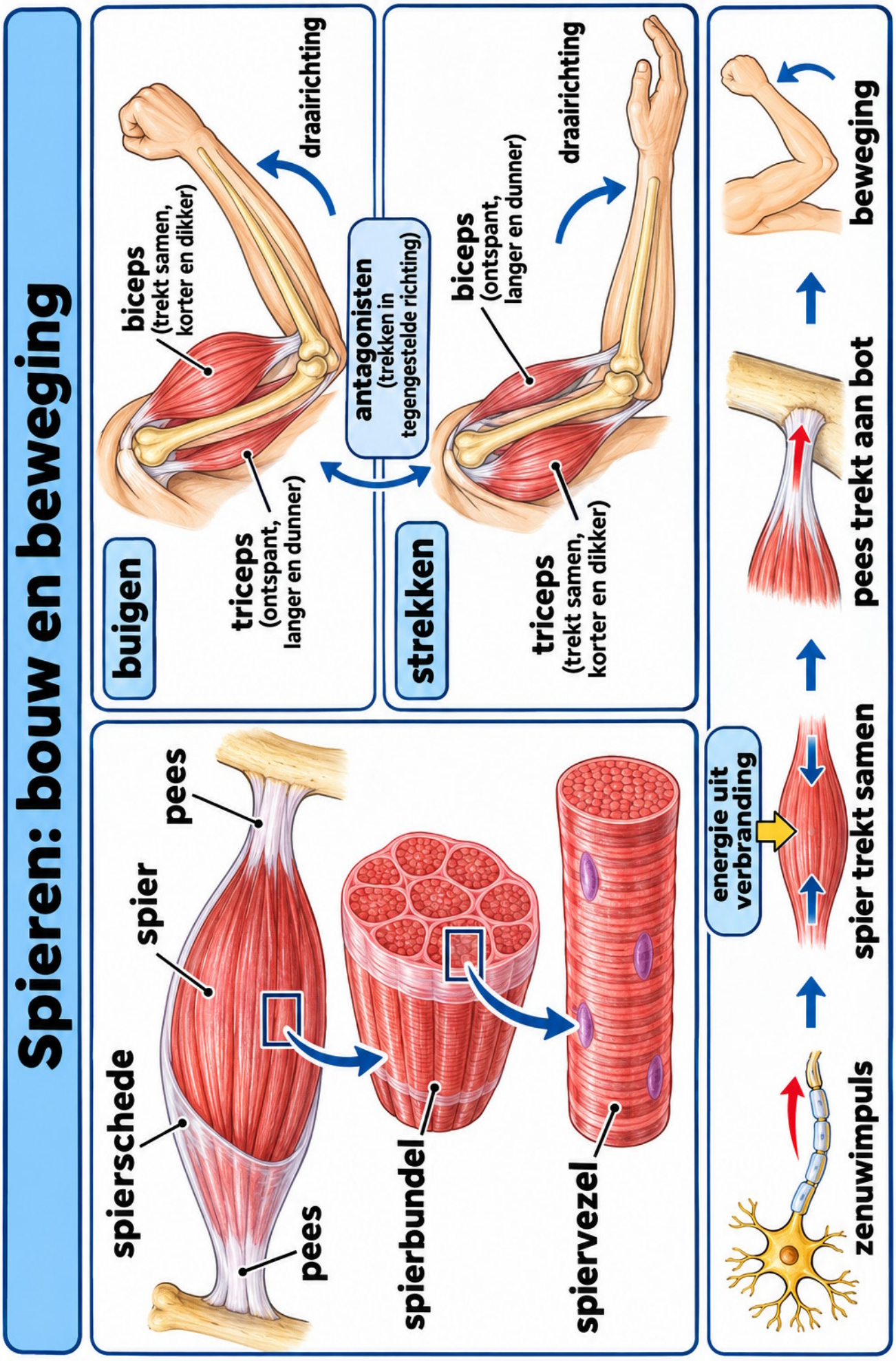
Basisstof 2, Kraakbeenweefsel en beenweefsel



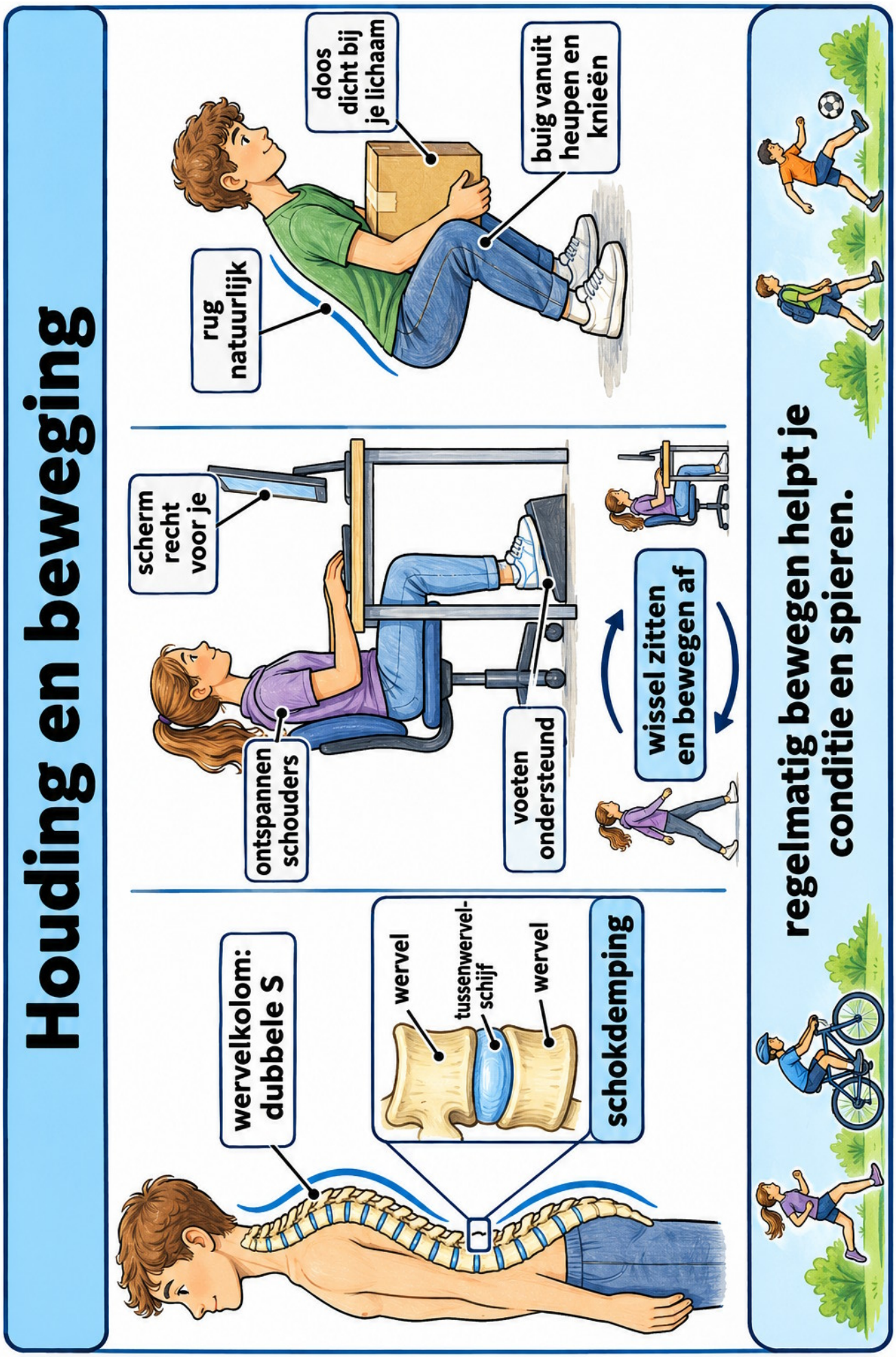
Basisstof 3, Beenverbindingen



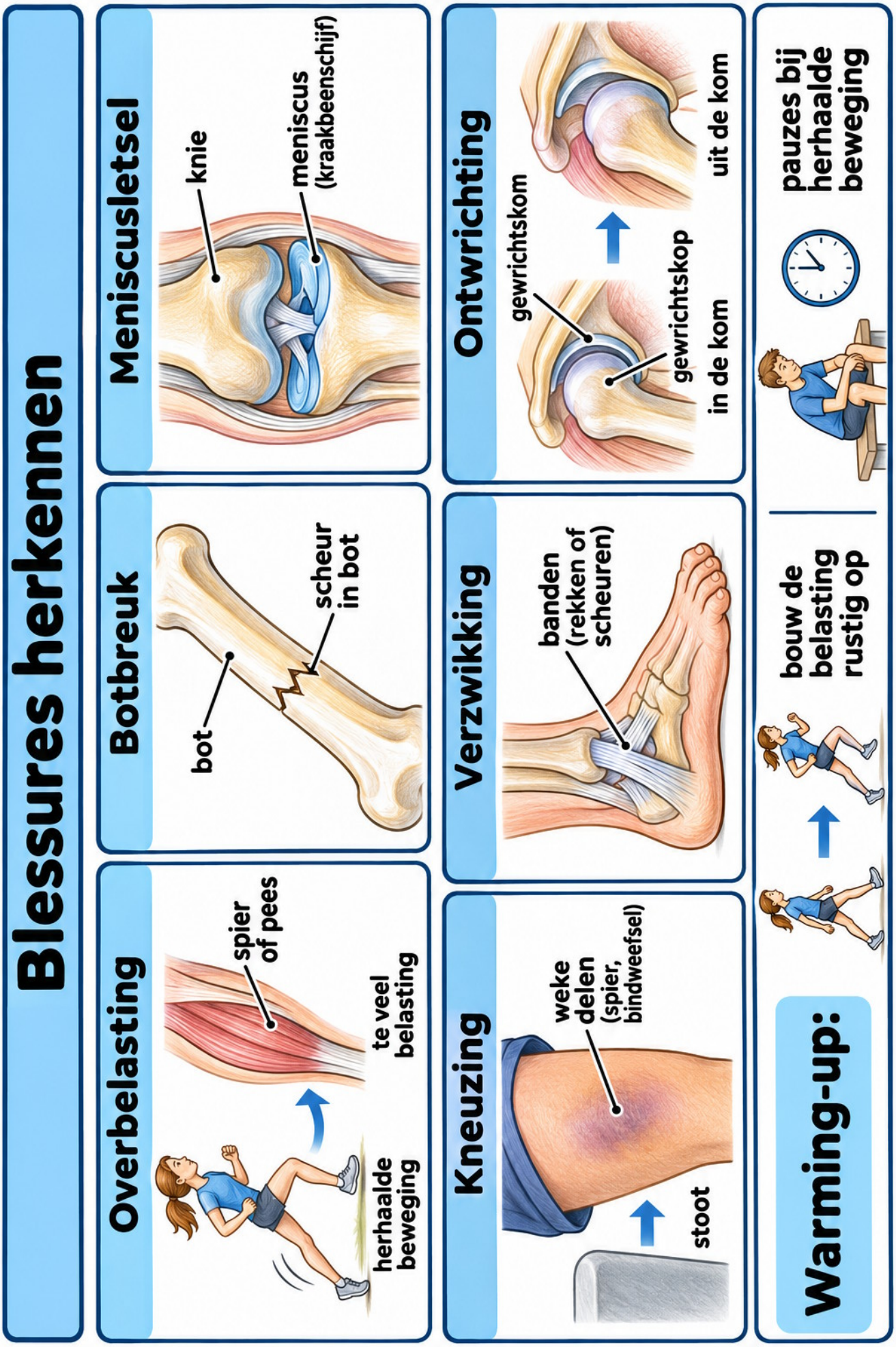
Basisstof 4, Spieren



Basisstof 5, Houding en blessures



Basisstof 6, Blessures

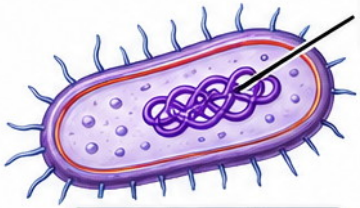


Organismen ordenen

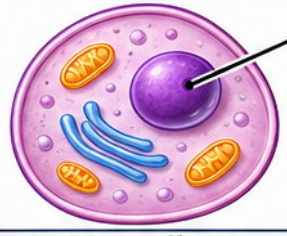
Wat is ordenen?

- Organismen indelen in groepen.
- Op basis van gemeenschappelijke kenmerken, vooral de bouw van de cellen.

Met of zonder celkern?



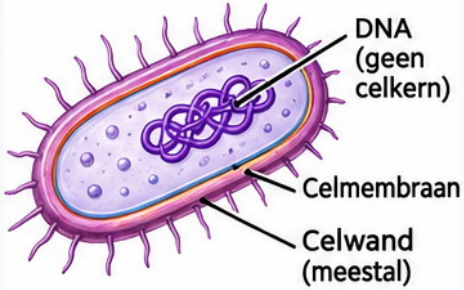
Zonder celkern
(bacterie)



Met celkern
(schimmel, plant, dier)

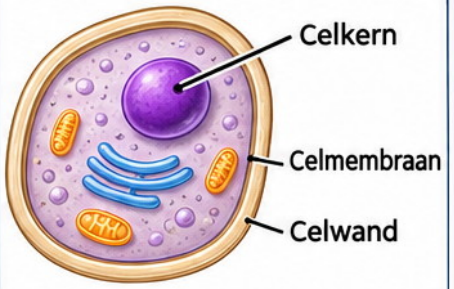
Vier groepen in het lesmodel (niet alle groepen van de moderne indeling)

Bacterie



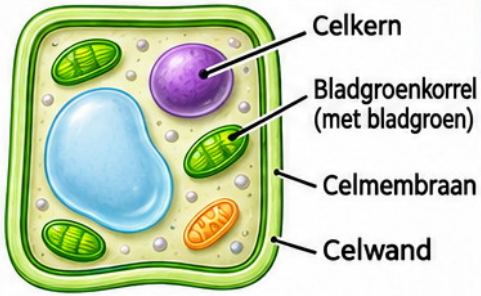
- DNA zonder celkern.
- Meestal een celwand.

Schimmelcel



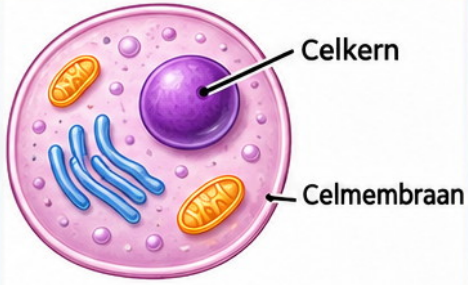
- Wel een celkern.
- Wel een celwand.
- Geen bladgroenkorels.

Groene plantencel



- Wel een celkern.
- Wel een celwand.
- Wel bladgroenkorrels.

Dierlijke cel

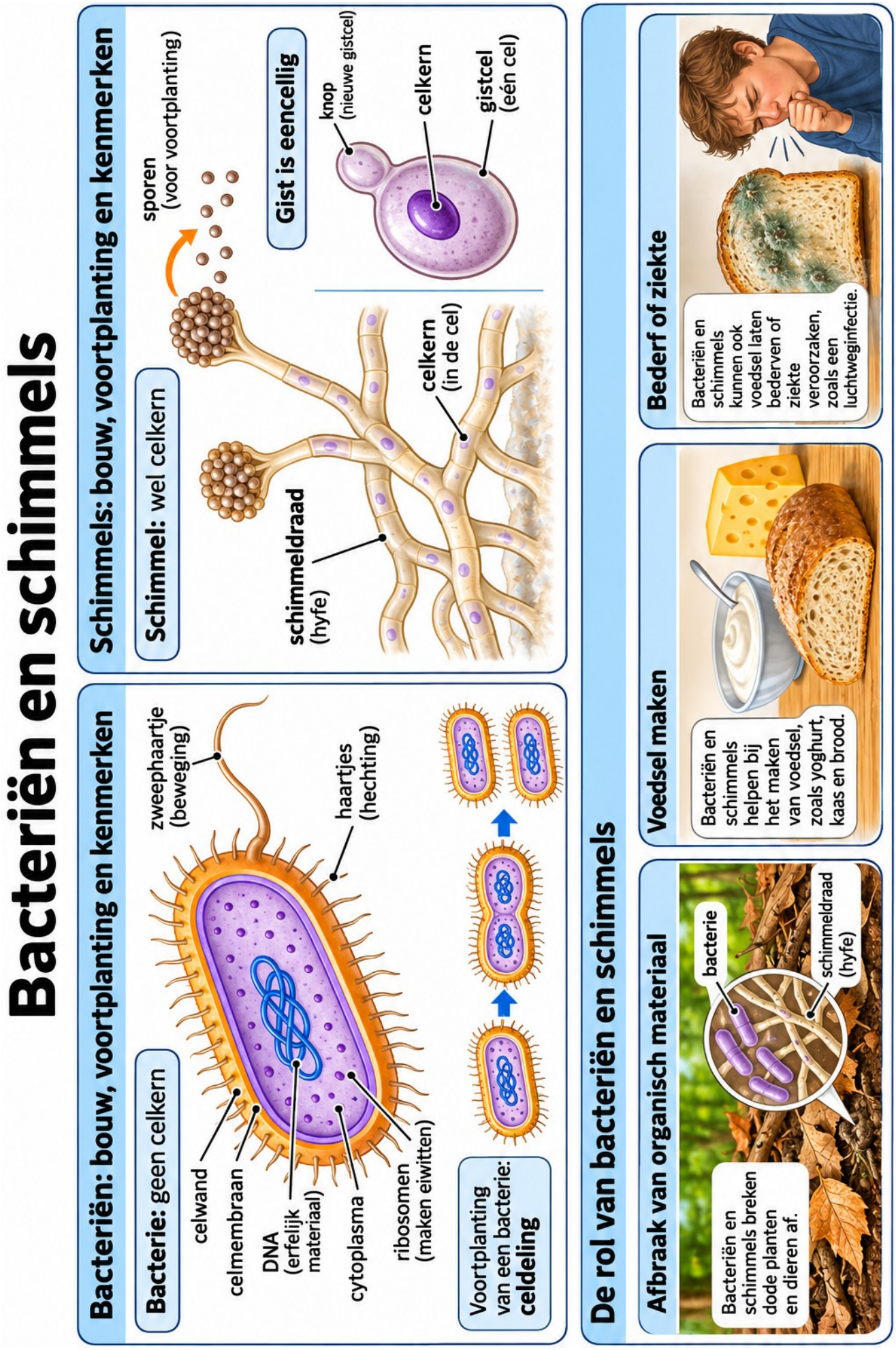


- Wel een celkern.
- Wel een celmembraan.
- Geen celwand.

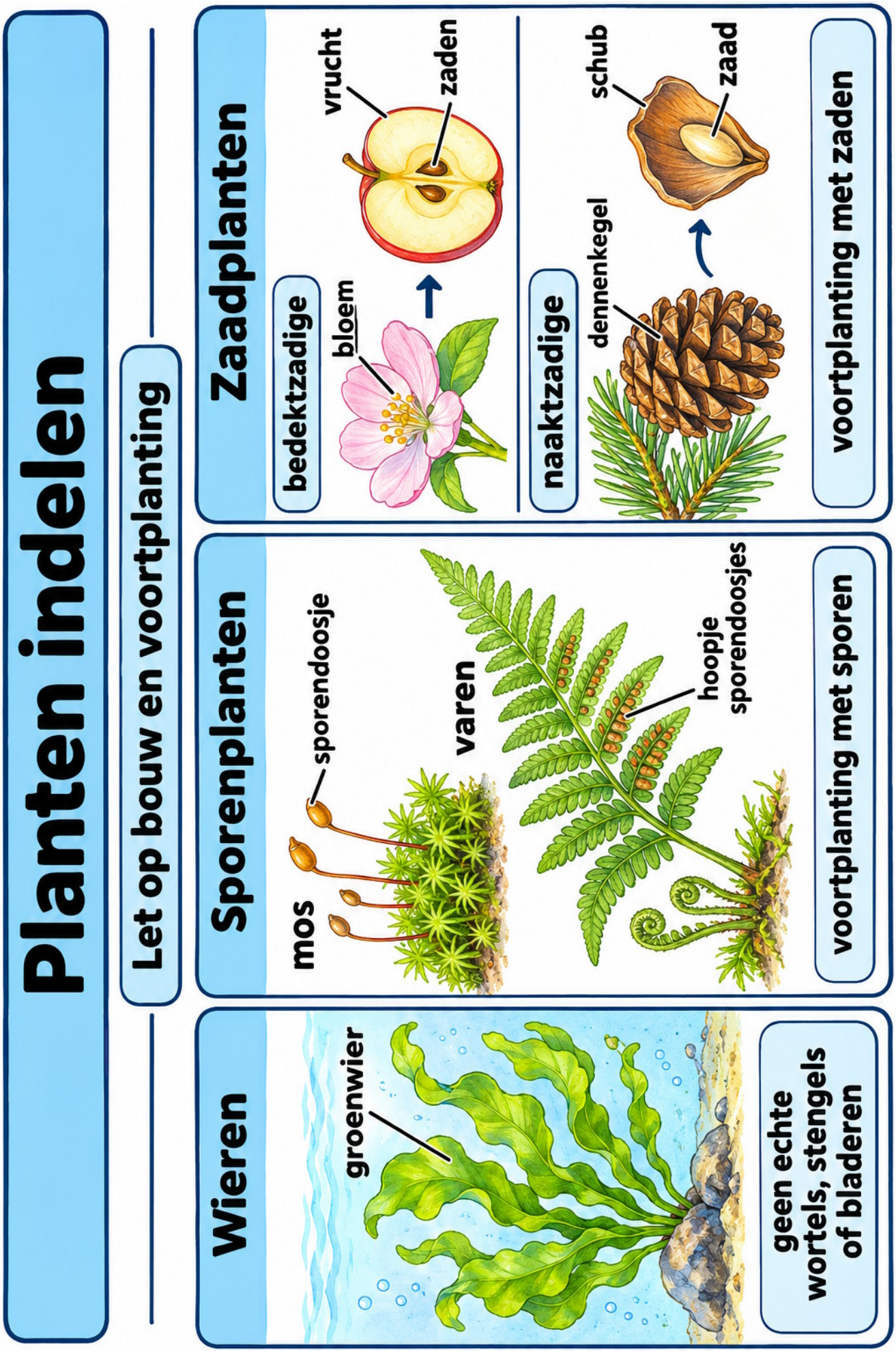
Indeling van groot naar klein

Rijk → **stam** → **klasse** → **orde** → **familie** → **geslacht** → **soort**

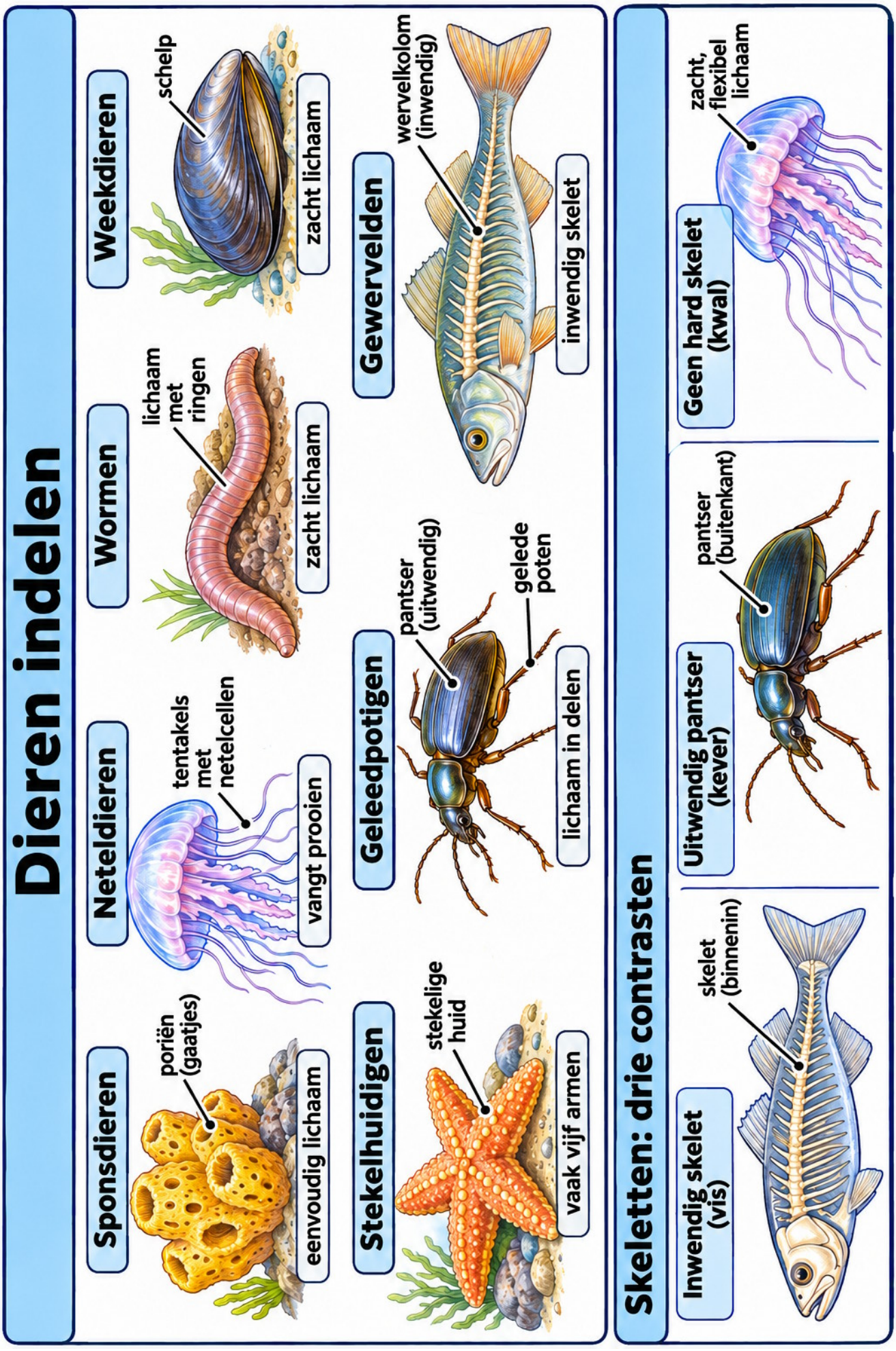
Basisstof 2 - Bacteriën en schimmels



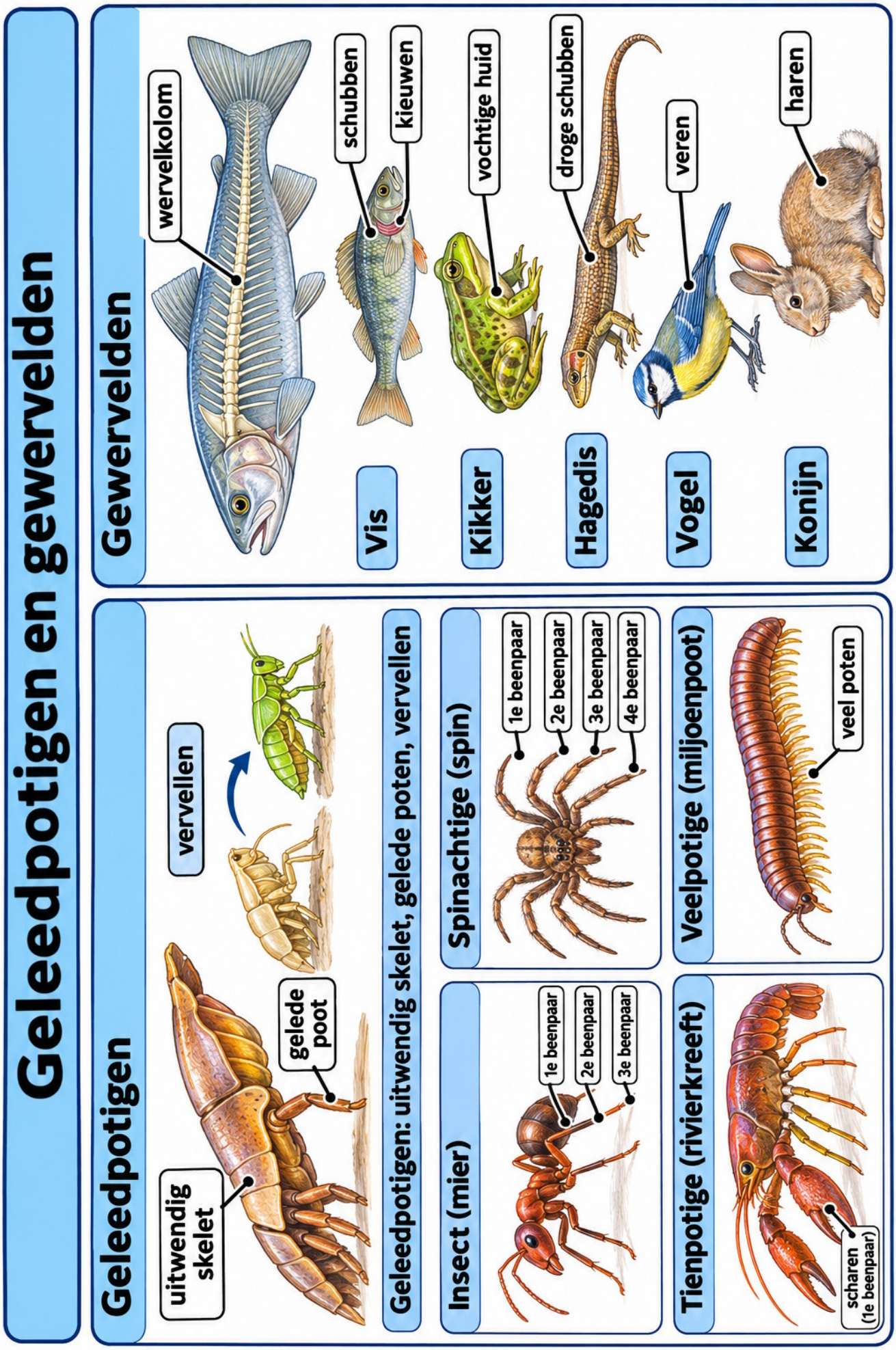
Basisstof 3 - Planten



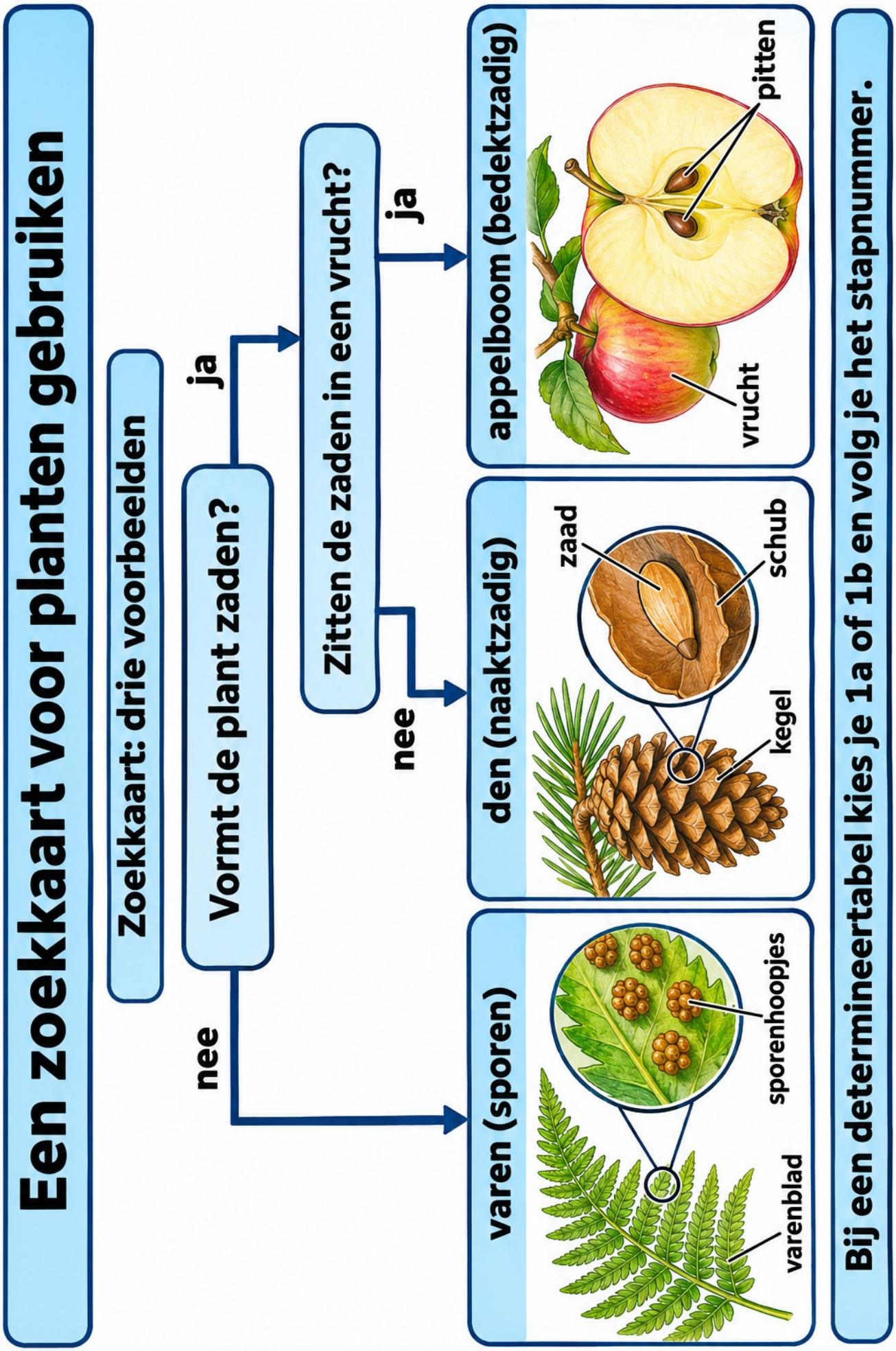
Basisstof 4 - Dieren



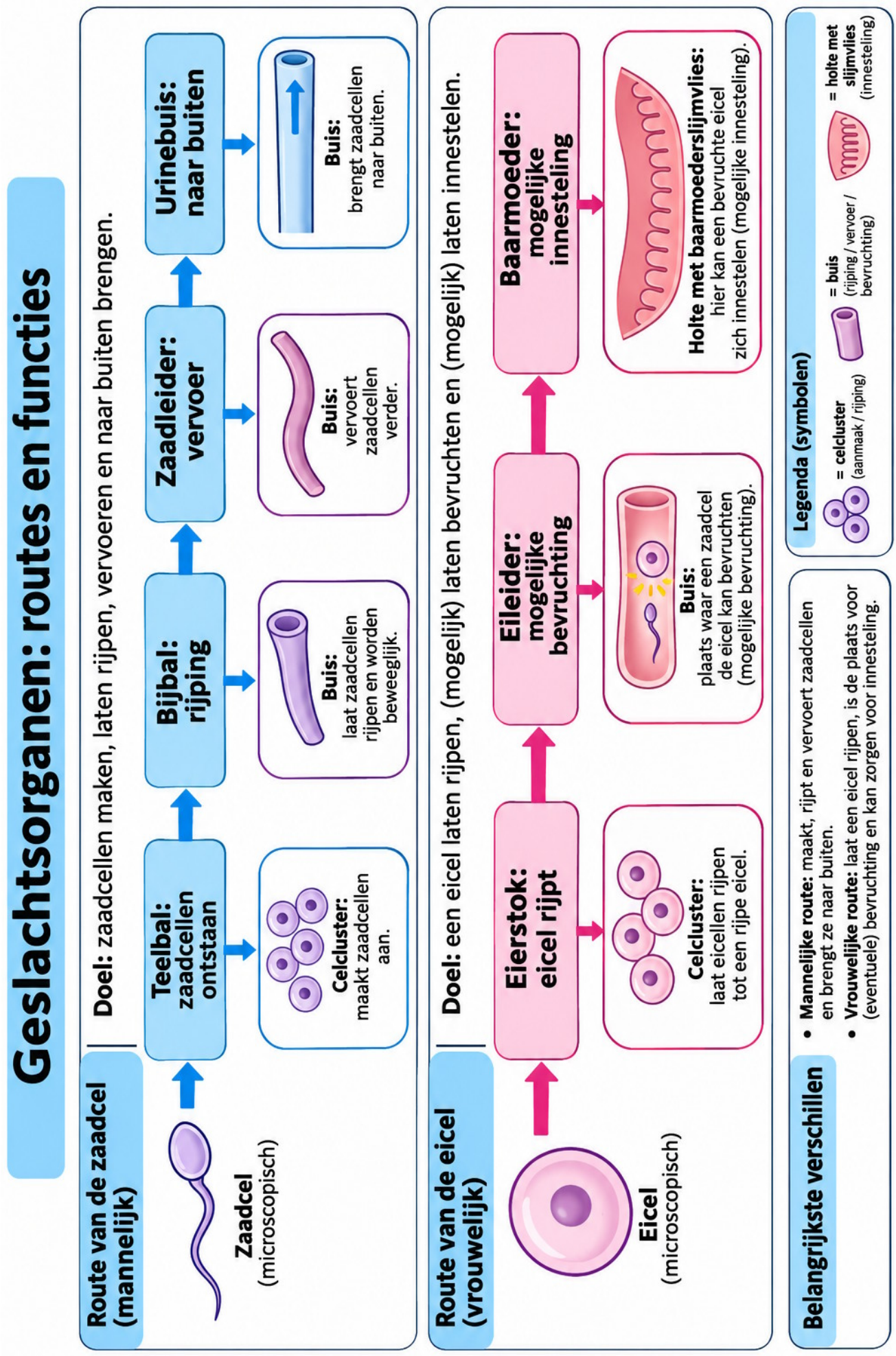
Basisstof 5 - Geleedpotigen en gewervelden



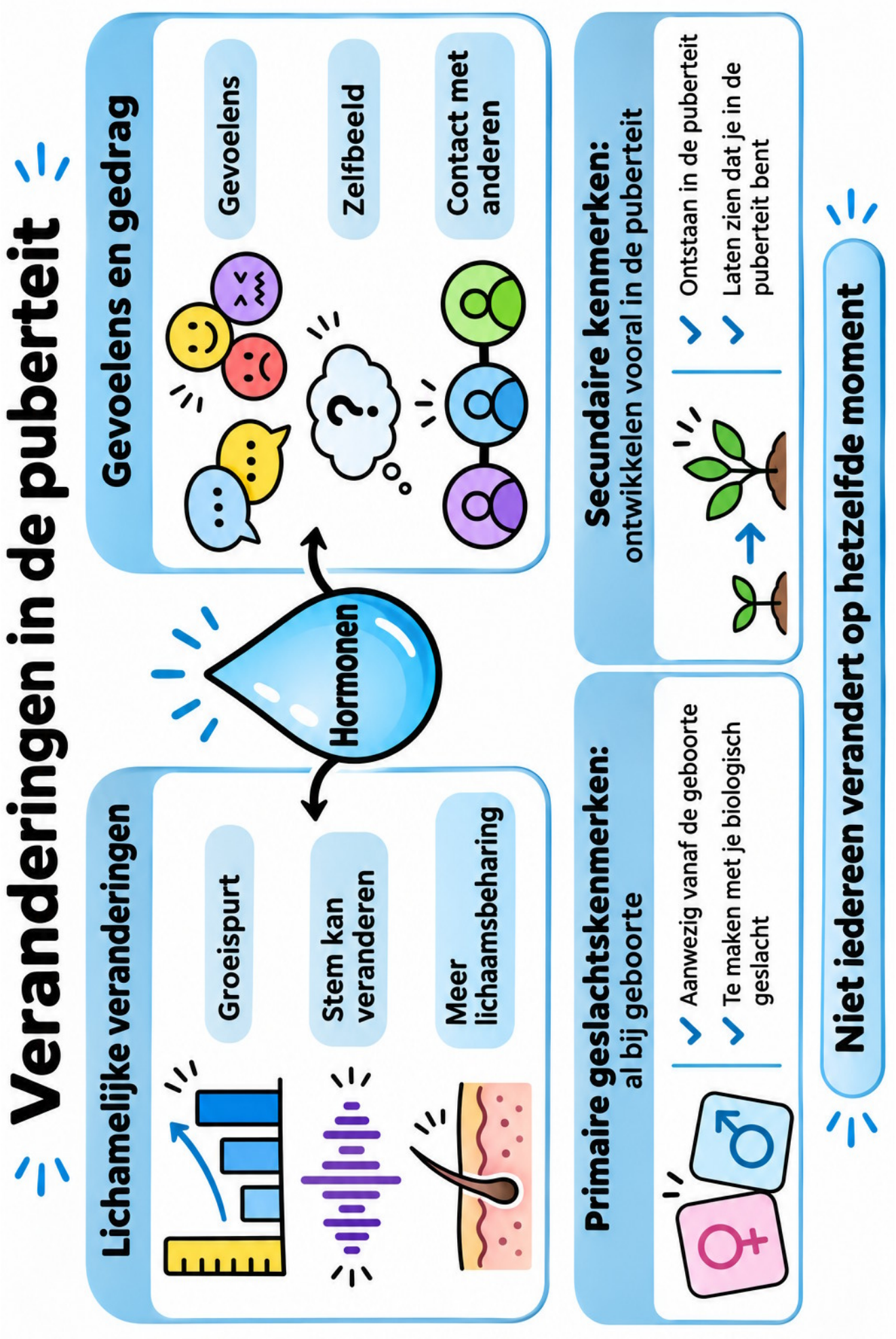
Basisstof 6 - Organismen determineren



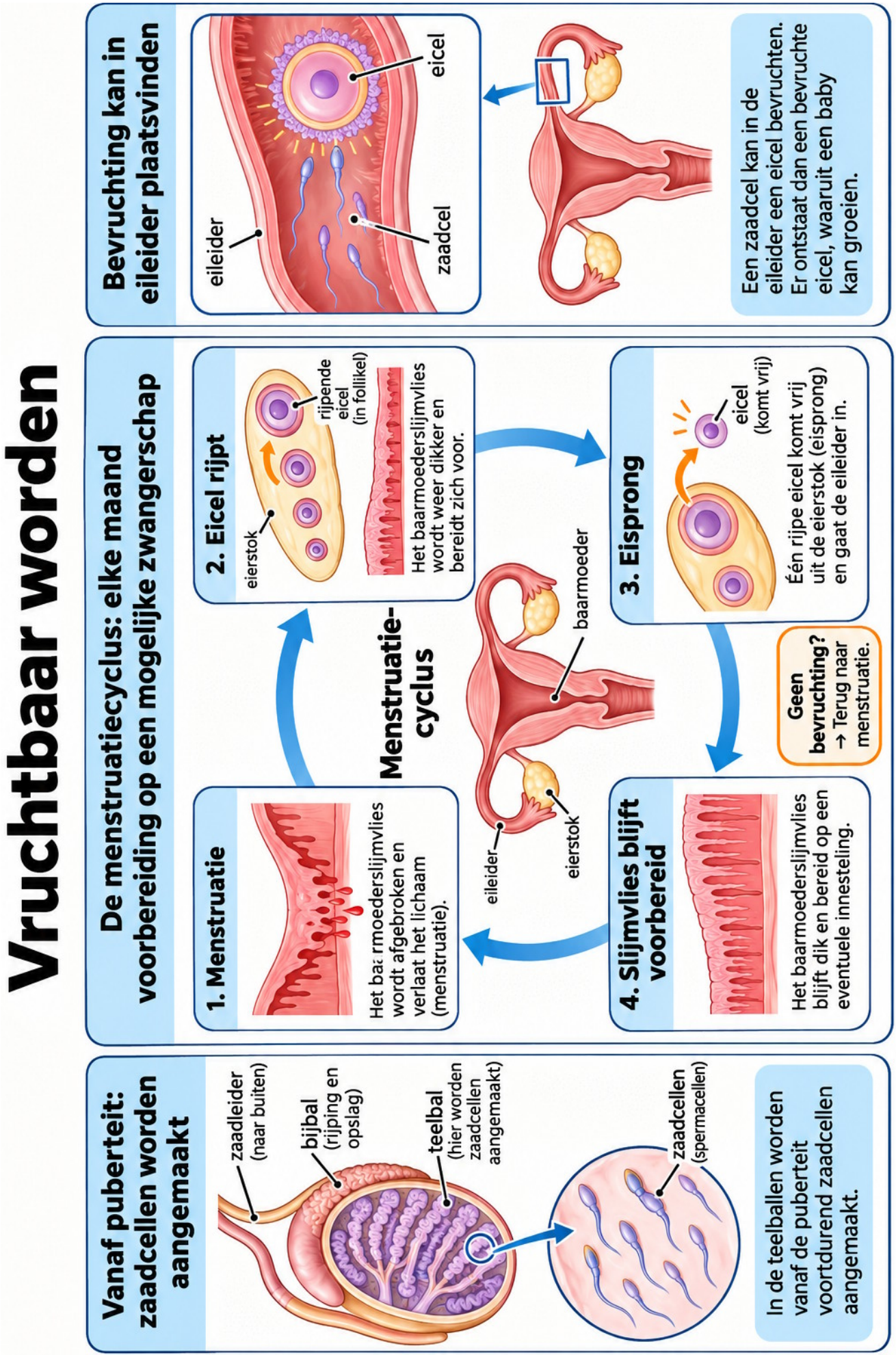
Basisstof 1 - Geslachtsorganen



Basisstof 2 - Veranderingen in de puberteit

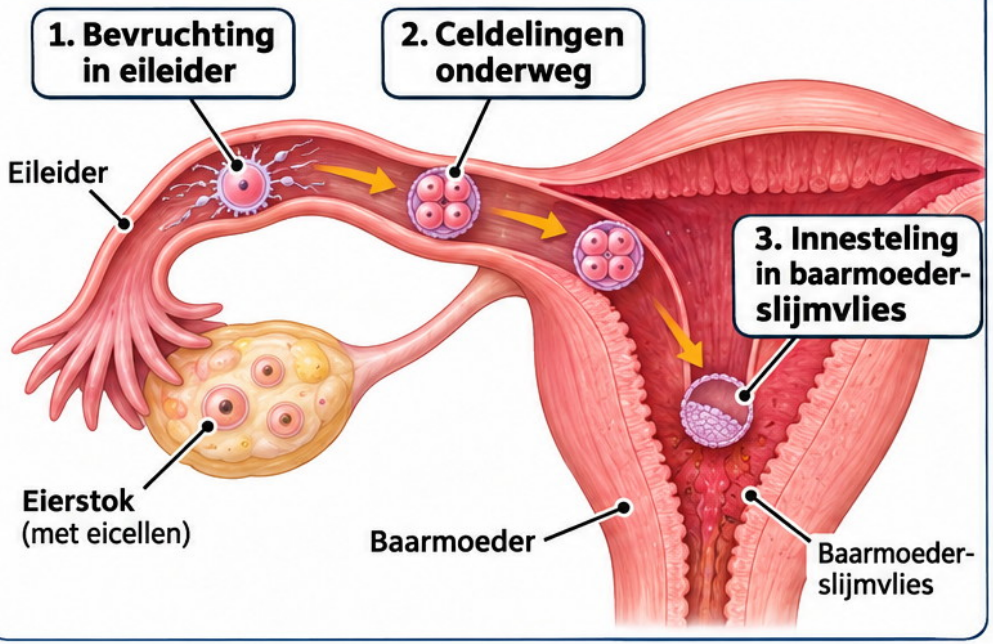


Basisstof 3 - Vruchtbaar worden

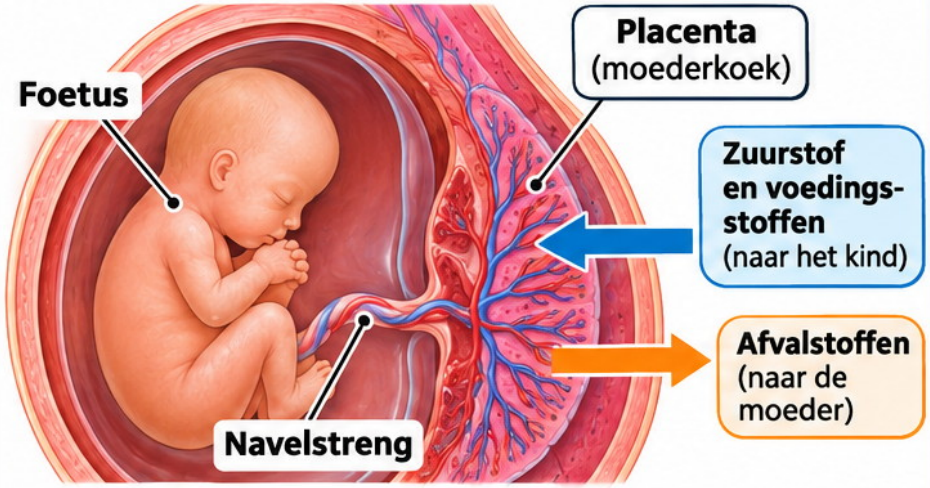


Zwanger worden

Van bevruchting tot innesteling



Zwangerschap: placenta en uitwisseling



Uitwisseling via de placenta: zuurstof en voedingsstoffen gaan naar het kind, afvalstoffen naar de moeder.
Het bloed van moeder en kind mengt niet.

Van embryo tot foetus

Embryo
- eerste weken
- organen ontstaan
- nog geen echte baby

Foetus
- vanaf ongeveer 8 weken
- organen werken
- groeit verder
- krijgt steeds meer de vorm van een baby

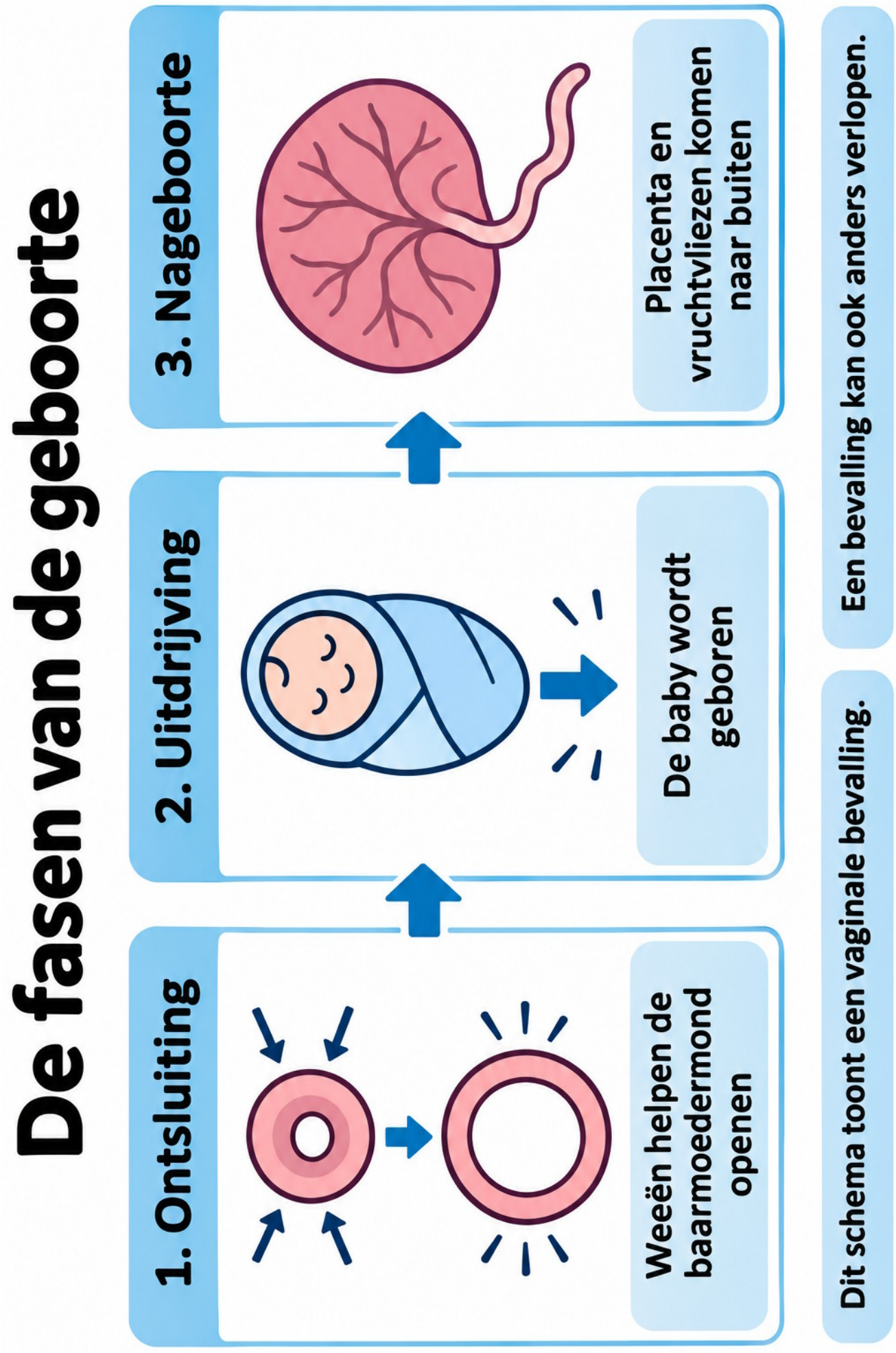
Embryo (ongeveer 4 weken)

Foetus (ongeveer 20 weken)

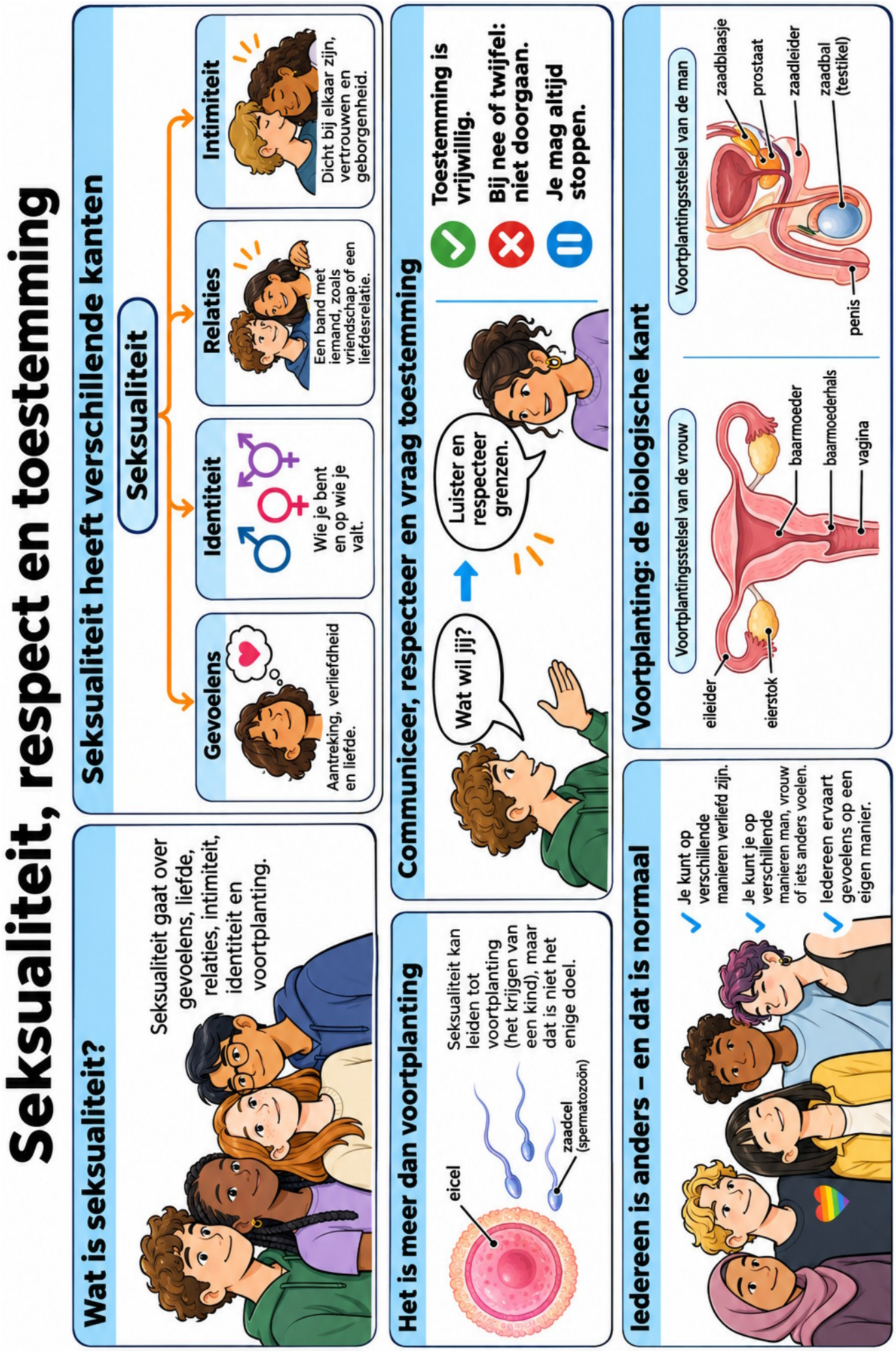
Kort samengevat

1. **Bevruchting:** een zaadcel bevrucht een eicel in de eileider.
2. **Celdelingen:** de bevruchte eicel deelt zich en groeit onderweg naar de baarmoeder.
3. **Innesteling:** het klompje cellen nestelt zich in het baarmoederslijmvlies.
4. **Zwangerschap:** het kind groeit in de baarmoeder, krijgt zuurstof en voedingsstoffen via de placenta en navelstreng, en geeft afvalstoffen aan de moeder af.

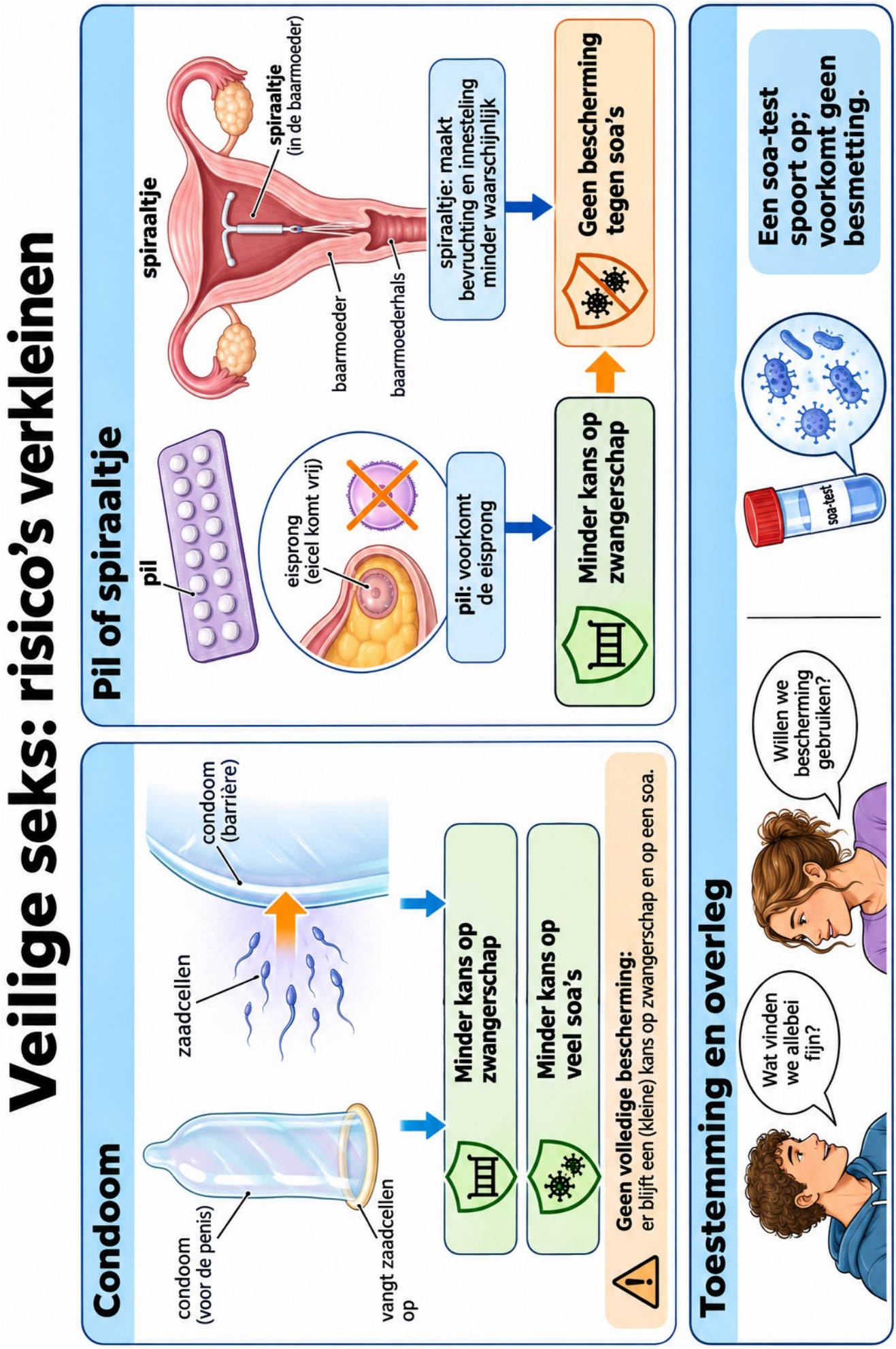
Basisstof 5 - Geboorte



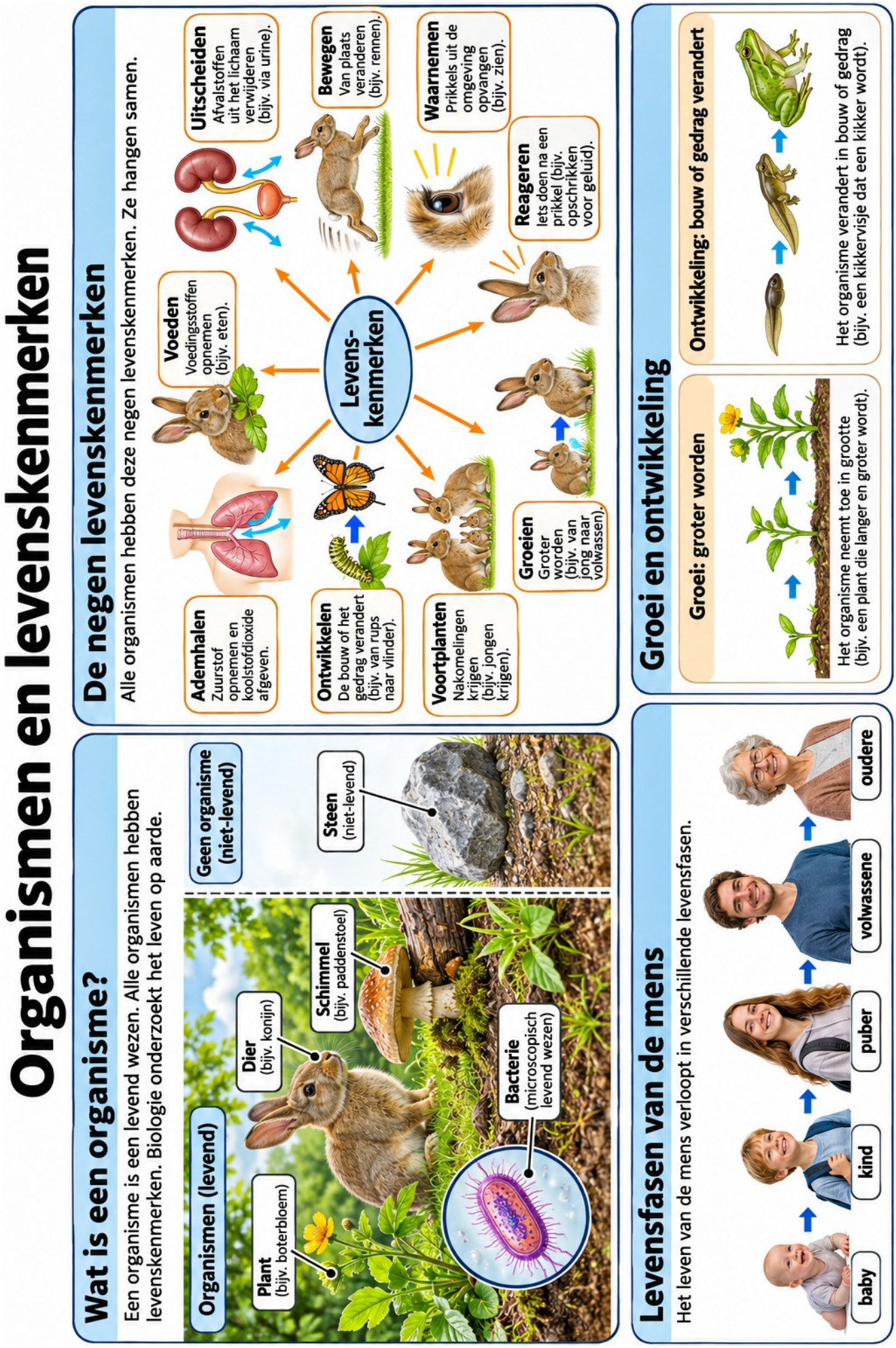
Basisstof 6 - Seksualiteit



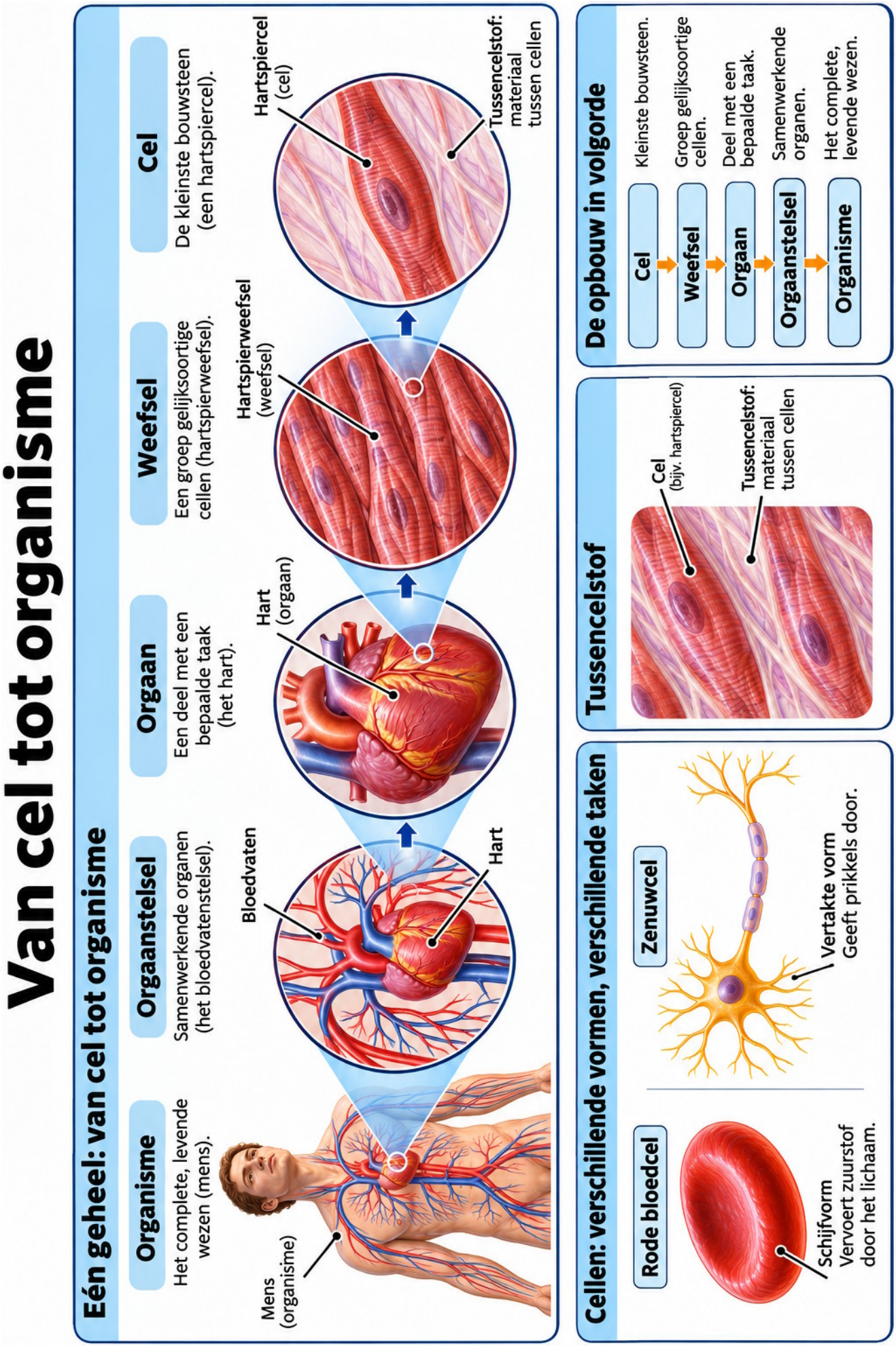
Basisstof 7 - Veilige seks



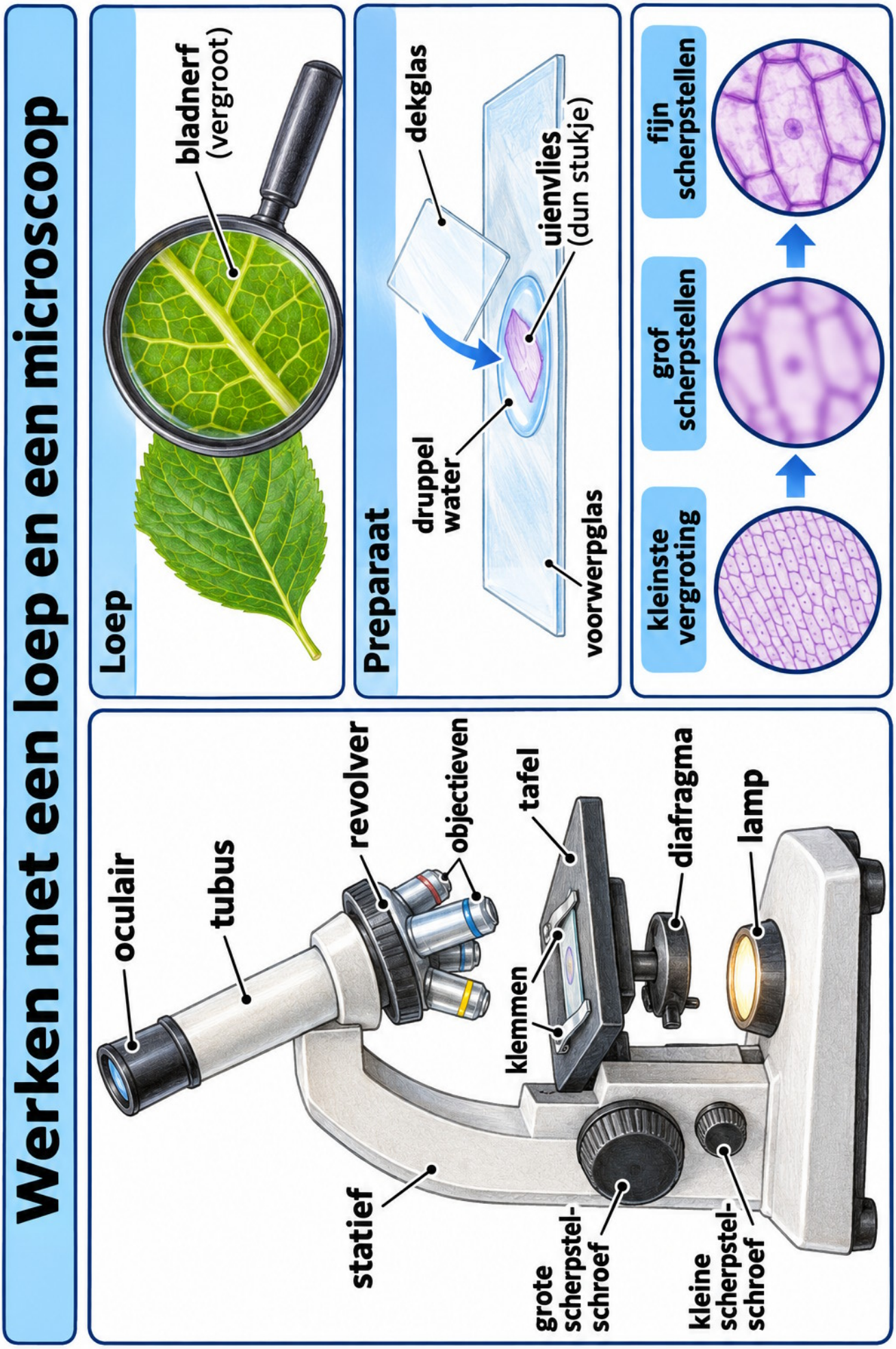
Basisstof 1 - Organismen



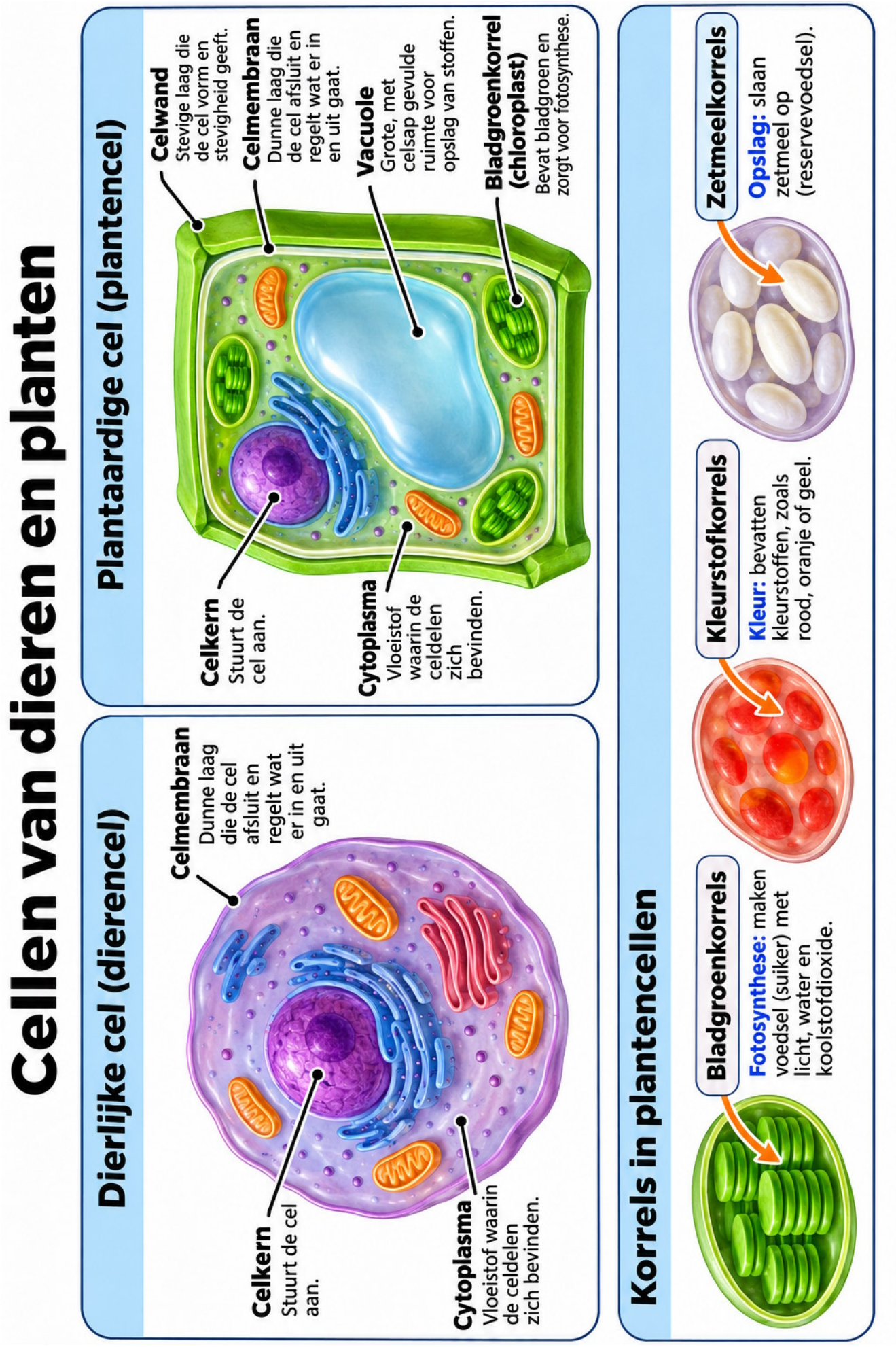
Basisstof 2 - De bouw van een organisme



Basisstof 3 - Werken met een loep en een microscoop



Basisstof 4 - Cellen van dieren en planten



Basisstof 5 - Groei en Ontwikkeling bij een Mens

Basisstof 6 - Biologisch onderzoek